

CXXC-finger protein 1 associates with FOXP3 to stabilize homeostasis and suppressive functions of regulatory T cells

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eLife Assessment

This study presents **important** findings on the role of CXXC-finger protein 1 in regulatory T cell gene regulation and function. The evidence supporting the authors' claims is **solid**, with mostly state-of-the-art technology, although the inclusion of more mechanistic insights would have strengthened the study. The work will be of relevance to immunologists interested in regulatory T cell biology and autoimmunity.

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Abstract

FOXP3-expressing regulatory T (T_{reg}) cells play a pivotal role in maintaining immune homeostasis and tolerance, with their activation being crucial for preventing various inflammatory responses. However, the mechanisms governing the epigenetic program in T_{reg} cells during their dynamic activation remain unclear. In this study, we demonstrate that CXXC finger protein 1 (CXXC1) interacts with the transcription factor FOXP3 and facilitates the regulation of target genes by modulating H3K4me3 deposition. *Cxxc1* deletion in T_{reg} cells

leads to severe inflammatory disease and spontaneous T-cell activation, with impaired immunosuppressive function. As a transcriptional regulator, CXXC1 promotes the expression of key T_{reg} functional markers under steady-state conditions, which are essential for the maintenance of T_{reg} cell homeostasis and their suppressive functions. Epigenetically, CXXC1 binds to the genomic regulatory regions of T_{reg} program genes in mouse T_{reg} cells, overlapping with FOXP3 binding sites. Given its critical role in T_{reg} cell homeostasis, CXXC1 presents itself as a promising therapeutic target for autoimmune diseases.

Introduction

Regulatory T (T_{reg}) cells are a distinct subset of CD4⁺ T cells that play a critical role in maintaining immune homeostasis and self-tolerance by suppressing excessive or aberrant immune responses to foreign or self-antigens^{1,2,3}. These cells can be further categorized into thymus-derived regulatory T cells (tT_{reg} cells), periphery-derived T_{reg} cells (pT_{reg} cells), and induced T_{reg} cells (iT_{reg} cells)⁴. They uniquely express the transcription factor FOXP3, a member of the forkhead winged-helix family, which is essential for T_{reg} cell lineage commitment and suppressive function^{5,6}. Deletion or mutation of the *Foxp3* gene leads to a range of immunological disorders, including allergies, immunopathology, and autoimmune diseases in both mice and humans^{7,8}.

It is well established that FOXP3 recruits various cofactors to form complexes that either promote or repress the expression of downstream genes, with histone and DNA modifications playing pivotal roles in this process. FOXP3 can activate or repress the transcription of key regulators of T_{reg} cell activation and function by recruiting the histone acetyltransferases (HATs) or histone deacetylases (HDACs)^{9,10}. Notably, FOXP3-bound sites exhibit significant enrichment of H3K27me3, a modification essential for FOXP3-mediated repressive chromatin remodeling under inflammatory conditions¹¹. However, the direct role of FOXP3 as a transcriptional activator through interactions with epigenetic regulators, particularly via modulation of H3K4 trimethylation, remains poorly documented.

In mammals, six proteins have been identified that catalyze H3K4 methylation. These proteins contain the SET domain and include MLL1 (KMT2A), MLL2 (KMT2B), MLL3 (KMT2C), MLL4 (KMT2D), SETD1A, and SETD1B^{12,13}. For example, MLL1 plays a critical role as an epigenetic regulator in T_{reg} cell activation and functional specialization¹⁴. Additionally, Placek *et al.* demonstrated that MLL4 is essential for T_{reg} cell development by catalyzing H3K4me1 at distant unbound enhancers through chromatin looping¹⁵. H3K4me3, which is enriched at the transcription start site (TSS) and the CpG island (CGI), converts chromatin into active euchromatin by recruiting activating factors¹⁶. CXXC finger protein 1 (CXXC1, also known as CFP1), which contains a SET1 interaction domain (SID), is required for binding to the histone H3K4 methyltransferases SETD1A and SETD1B¹⁷. Previous studies have shown that CXXC1 regulates promoter patterns during T cell maturation, GM-CSF-derived macrophage phagocytosis, TH17 cell differentiation, and the function of ILC3 cells with aging by modulating H3K4me3 modification^{18–21}. Despite the well-documented role of CXXC1 in various immune effector cells, its role in T_{reg} cells remains unclear.

Here, we demonstrate that CXXC1 interacts with FOXP3 and enhances the expression of FOXP3 target genes. T_{reg} cell-specific deletion of *Cxxc1* triggers systemic autoimmunity, accompanied by multiorgan inflammation, ultimately resulting in early-onset fatal inflammatory disease in mice. *Cxxc1*-deficient T_{reg} cells exhibit a significant disadvantage in proliferation and homeostasis, even in non-inflammatory mice where coexisting wild-type (WT) T_{reg} cells were present. Moreover, *Cxxc1*-deficient T_{reg} cells display intrinsic defects in the expression of key suppression molecules,

including CTLA-4, CD25, ICOS, and GITR. Consistent with these findings, mechanistic studies strongly suggest that CXXC1 functions as an essential cofactor of FOXP3, playing a crucial role in sustaining H3K4me3 modifications at key T_{reg} cell genes.

Materials and methods

Mice

All mice used in this study were bred for a minimum of seven generations on a C57BL/6 background. Mouse experiments, or cells from mice of the same genotype, compared littermates or age-matched control animals. The *Cxxc1*^{fl/fl} mouse strain has been previously described²². The *Foxp3*^{cre} mice (JAX,016959) were generously provided by Bin Li (Shanghai Jiao Tong University School of Medicine, Shanghai, China). CD45.1 (NM-KI-210226) mice were purchased from the Nanjing Biomedical Research Institute of Nanjing University. *Rag1*^{-/-} mice (stock#T004753) were purchased from GemPharmatech. 2D2 (MOG35-55-specific TCR transgenic) mice were graciously supplied by Prof. Linrong Lu (Zhejiang University School of Medicine, Hangzhou, Zhejiang, China). *Foxp3*^{cre} (WT) and *Foxp3*^{cre} *Cxxc1*^{fl/fl} (cKO) were utilized at 3 weeks old unless otherwise stated. All mice were housed in the Zhejiang University Laboratory Animal Center under specific pathogen-free conditions, and all animal experimental procedures were approved by the Zhejiang University Animal Care and Use Committee (approval no.ZJU20230246).

Cell culture

HEK 293T cells were obtained from ATCC, and Plat E cells were kindly provided by Prof. Xiaolong Liu (Shanghai Institutes for Biological Sciences). Both cell lines were cultured in Dulbecco-modified Eagle medium (DMEM) containing 10% (v/v) fetal bovine serum (FBS), supplemented with 1% penicillin/streptomycin.

Immunofluorescence (IF) Staining

As previously described²³, coverslips were treated with a 0.01% poly-L-lysine solution (P4707; Sigma) for 10 minutes, air-dried, and then coated with CD4⁺ YFP⁺ T_{reg} cells. The cells were then fixed in 4% formaldehyde for 15 minutes at room temperature, permeabilized with 0.2% Triton X-100, and blocked with 1% BSA. Antibodies against CXXC1 (ab198977; Abcam) and FOXP3 (17-5773-82; Invitrogen) were diluted in Image iT FX signal enhancer (I3693; Invitrogen) and incubated with cells overnight at 4 °C. After washing with phosphate-buffered saline (PBS), the cells were incubated with a goat anti-rabbit antibody Alexa Fluor 594 (1:250;10015289; Invitrogen) secondary antibody and stained with DAPI (200 ng/mL; D523; Dojindo). Slides were washed with PBS and sealed with an antifade solution (P36934; Invitrogen) before imaging with an Olympus FV3000 fluorescence microscope. The images were visualized using the FV31-SW software.

Co-immunoprecipitation and western blot

Harvest the appropriate transfected cell lines and primary cells from culture and wash them with ice-cold PBS. Lyse the cells in NETN300 buffer (300 mM NaCl, 0.5 mM EDTA, 0.5% (v/v) NP-40, 20 mM Tris-HCl pH 8.0) supplemented with a protease inhibitor (1:100, P8340, Sigma-Aldrich) and PMSF (1 mM) on ice for 10 minutes. Take a small portion of the whole-cell lysate as input, and incubate the remaining lysate with either Anti-FLAG M2 Beads (M8823; Sigma) or Anti-HA Beads (HY-K0201; MCE) on a rotator at 4°C overnight. For the endogenous Co-IP assay targeting CXXC1 and FOXP3, incubate the cell lysate with protein G magnetic beads along with anti-CXXC1 (ab198977; Abcam) or anti-FOXP3 (14-4774-82; Invitrogen) antibodies on a rotator at 4°C overnight. Wash the beads three times with IP buffer (100 mM NaCl, 0.5 mM EDTA, 0.5% (v/v) NP-40, 20 mM Tris-HCl pH 8.0) to remove non-specific binding. Boil the washed beads with 1× Laemmli sample buffer (1610747; Bio-Rad) to elute the bound proteins. Separate the denatured proteins by SDS-

PAGE. Transfer the separated proteins onto PVDF membranes (Millipore) for immunoblotting. Immunoblot the PVDF membranes (IPVH00010) with the following antibodies: anti-CXXC1 (1:1000; ab198977; Abcam), anti-FOXP3 (1:500; 14-7979-80; Invitrogen), anti-FLAG (1:1000; 14793; Cell Signaling Technology), anti-HA (1:1000; 3724S; Cell Signaling Technology). Detect the immunoblotted proteins using a secondary HRP-conjugated goat anti-rabbit antibody (1:1000; HA1001-100; Huabio) and visualize the bands using an appropriate detection method.

Flow cytometry

Spleen and lymph nodes were harvested from 3-week-old mice. Tissues were placed in a dish containing PBS with 2% fetal bovine serum (FBS). The tissues were minced into small pieces and then filtered through a 70 μ M nylon filter to obtain a single-cell suspension. For surface marker analysis, cells were incubated for 15 minutes with purified anti-mouse CD16/32 antibody (101320; BioLegend) to block Fc receptors. After blocking, cells were stained with the indicated antibodies for surface markers. To determine cytokine expression, cells were stimulated for 4 hours at 37°C with phorbol12-myristate13-acetate (50ng/mL; S1819; Beyotime), ionomycin (1mg/mL; S1672; Beyotime), and brefeldin A (BFA; 00-4506-51; Invitrogen). After stimulation, cells were labeled with a fixable viability dye and surface markers. Cells were then fixed and permeabilized according to the manufacturer's instructions (00-8222-49; Invitrogen). For transcription factor staining, samples were fixed using the FOXP3/Transcription Factor Staining Buffer Set (00-5523-00; Invitrogen). Flow cytometry was performed using a BD Fortessa (BD Biosciences). Flow cytometry data were acquired and analyzed using FlowJo software.

The following antibodies were purchased from Invitrogen or BioLegend: Zombie Violet fixable viability (423113), Zombie NIR fixable (423105), CD25(PC61), CD8 α (53-6.7), CD62L(MEL-14), PD-1(J43), CD44(IM7), IL-17A(TC1118H10), KL-RG1(2F1), CD4 (GK1.5), TCR β (H57-597), IFN- γ (XMG1.2), FOXP3(FJK-16s), CTLA-4(UC10-4B9), ICOS(15F9), GITR(DTA-1), LAG3(C9B7W), TCRV β 11(RR3-15), CD45RB(C363-16A) CD69 (H1.2F3), MHCII(M5/114.15.2), CD80(16-10A1), T-bet(eBio4B10), IKZF4(ESB7C2), IL-4(11B11), CCR7(4B12), CD73(TY/11.8).

Real-time PCR

Total RNA was extracted from T_{reg} cells using the RNAiso Plus(9109; Takara) reagent according to the manufacturer's instructions, and cDNA synthesis was performed using the Prime Script RT Reagent Kit (Takara). TB Green Premix Ex Taq (RR420A; Takara) was used for quantitative real-time PCR (qPCR). The expression levels of target mRNA were normalized to the level of β -actin expression. The primers for qPCR are as follows:

Cxhc1 qPCR Forward: ATCCGGAATGGTACTGTGCG

Cxhc1 qPCR Reverse: CTGTGGAGAAGATTGTGGG

β -actin qPCR Forward: CTGTCCCTGTATGCCTCTG

β -actin qPCR Reverse: ATGTACGCACGATTTCC

CD4⁺T and YFP⁺ T_{reg} cells adoptive transfer in

Experimental autoimmune encephalomyelitis (EAE)

CD4⁺YFP⁺ T_{reg} cells from *Foxp3*^{cre} and *Foxp3*^{cre} *Cxhc1*^{fl/fl} mice were enriched using the Mouse CD4 T Cell Isolation Kit (480005; Biolegend) and then sorted using the BD Aria II flow cytometer. Naïve CD4⁺T cells from 2D2 (MOG35-55-specific TCR transgenic) mice were isolated using the Mouse CD4 Naïve T cell Isolation Kit480039 (480039; Biolegend). As previously described²⁴, 2D2 naïve CD4⁺

T cells alone (5×10^5 per mouse), or 2D2 naïve $CD4^+$ T cells (5×10^5 per mouse) together with WT or *Foxp3^{cre}Cxhc1^{fl/fl}* T_{reg} cells (2×10^5 per mouse), were transferred into *Rag1^{-/-}* mice via the tail vein. One day after cell transfer, the recipient mice were inoculated subcutaneously (s.c.) with 200 µg MOG35-55 peptide (MEVGWYRSPFSRVVHLYRNGK; GenemeSynthesis) emulsified in complete Freund's adjuvant (CFA) (F5506; Sigma). Intravenous administration of 200 ng of Pertussis toxin (181; List Biological Laboratories) was performed on days 0 and 2 after peptide inoculation. The severity of EAE was monitored and blindly graded using a clinical score from 0 to 5: 0, no clinical signs; 1, limp tail; 2, paraparesis (weakness, incomplete paralysis of one or two hind limbs); 3, paraplegia (complete paralysis of two hind limbs); 4, paraplegia with forelimb weakness or paralysis; 5, dying or death.

Isolation lymphocytes from the central nervous system (CNS)

On day 14 after EAE induction, mice were perfused with transcardially administered PBS to eliminate contaminating blood cells in the central nervous system (CNS). The forebrain and cerebellum were dissected to expose the spinal cord, which was then carefully removed from the spinal canal. The fresh spinal cord was harvested and cut into 2 mm pieces. The CNS tissue pieces were homogenized using a syringe and passed through a 70 µm cell strainer to obtain a single-cell suspension. The single-cell suspension was digested with collagenase D (2 µg/ml; 11088858001; Roche) and deoxyribonuclease I (DNase I; 1 µg/ml; DN25; Sigma-Aldrich) at 37°C for 20 minutes under rotation. After digestion, the cell suspension was centrifuged to pellet the cells. The cell pellets were resuspended in 40% Percoll and layered onto a discontinuous Percoll gradient. Centrifugation at 80% Percoll allowed for the separation of cells at the 40–80% Percoll interface, which were collected as CNS mononuclear cells. The collected CNS mononuclear cells were washed with PBS to remove any remaining Percoll. CNS mononuclear cells were stimulated for 4 hours with PMA and ionomycin in the presence of Brefeldin A to induce cytokine production. After stimulation, cells were fixed, rendered permeable, and stained with appropriate antibodies for intracellular cytokine detection.

Histological analyses

The lungs, skin, liver, and colon were excised from three-week-old mice. Prior to histological analysis, the samples were fixed in formalin, embedded in paraffin, and stained with hematoxylin and eosin. For CNS histology, spinal cords were fixed in 4% paraformaldehyde, paraffin-embedded, sectioned, and stained with Luxol Fast Blue and hematoxylin and eosin (H&E). To examine colon histology, colons from *Rag1^{-/-}* hosts were similarly processed and stained with H&E.

Adoptive transfer colitis model

Colitis was induced following the protocol described²⁵. In brief, $CD4^+$ YFP⁺ T_{reg} cells were isolated from 3-week-old $CD45.2^+$ *Foxp3^{cre}Cxhc1^{fl/fl}* and $CD45.2^+$ *Foxp3^{cre}* mice. A total of 2×10^5 T_{reg} cells from each group were mixed with 4×10^5 T_{eff} cells ($CD45.1^+CD4^+CD45RB^{hi}$) sorted from $CD45.1^+$ mice and transferred into the *Rag1^{-/-}* mice via intraperitoneal injection. T_{eff} cells alone were transferred as a control group. Mouse body weight was measured weekly post-adoptive transfer. The percentage change in body weight was calculated by comparing the current weight with the initial weight on day 0. Mice were euthanized when any had reached 80% of their initial body weight. The large intestines were sectioned into 4 µm thick slices and stained with hematoxylin.

In vitro T_{reg} suppression assay

Naïve $CD4^+$ T cells isolated from WT mice were labeled with CFSE (C34554; Invitrogen). $CD4^+$ YFP⁺ T_{reg} cells from *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice were cultured with naïve $CD4^+$ T cells (1×10^5 cells) at various ratios in the presence of 2 µg/mL anti-CD3 (16-0031-85; Invitrogen) and 3 µg/mL

anti-CD28(16-0281-85; Invitrogen). On day 3, cells were analyzed by flow cytometry.

CUT&Tag

CUT&Tag assays of CD4⁺YFP⁺ T_{reg} cells were conducted as previously described²⁶. Briefly, approximately 1x10⁵ single cells were carefully pipetted into wash buffer twice. The pelleted cells were resuspended in wash buffer, activated concanavalin (BP531; Bangs Laboratories), and incubated for 15 minutes at room temperature. Cells bound to the beads were resuspended in Dig-Wash Buffer and incubated with a 1:50 dilution of primary antibodies (rabbit anti-H3K4me3, Active Motif, 39016; rabbit anti-CXXC1, abcam, ab198977; normal IgG, Cell Signaling, 2729) at 4 °C overnight. The beads were incubated with a secondary antibody (goat anti-rabbit IgG; SAB3700883; Sigma-Aldrich) diluted 1:100 in Dig-Wash buffer for 60 minutes at room temperature. Cells were treated with Hyperactive pG-Tn5 Transposase (S602; Vazyme) diluted in Dig-300 Buffer for 1 hour at room temperature. The cells were subsequently resuspended in Tagmentation buffer (10mM MgCl₂ in Dig-300 Buffer) and incubated at 37°C for 1 hour. To halt tagmentation, 10μl was spiked with 0.5M EDTA, 3μl with 10%SDS, and 3μl with 20mg/ml Proteinase K and incubated at 55°C for 1h. DNA library amplification was performed according to the manufacturer's instructions and purified using VAHTS DNA Clean Beads (N411; Vazyme). Libraries were sequenced on the Illumina NovaSeq platform (Annoroad Gene Technology).

CUT&Tag and ChIP-seq data analysis

FOXP3 ChIP-seq data was obtained from GSE121279. H3K27me3 ChIP-seq data was obtained from GSE14254. CUT&Tag and ChIP-seq reads were trimmed to 50 bp and aligned against the mouse genome build mm9 using Bowtie2(v2.3.4.1) with default parameters. All PCR duplicates and unmapped reads were removed. Peak calling was performed using MACS2(v2.1.1.20160309) and signal tracks for each sample were generated using the 'wigToBigWig' utility of UCSC. We classified the H3k4me3 peaks around TSSs into three groups: broad (>5 kb), medium (1-5kb), and narrow (<1 kb). The top 5% of the widest peaks were considered as broad peaks. The average intensity profiles were generated using deepTools (v2.5.4). Motif analysis was performed using the 'findmotifsGenome.pl' command in Homer2 package. Epigenetic factors were identified using the Epigenetic Factor Database(<https://epifactors.autosome.org/>) and then screened for those that exclusively regulate the expression of their target genes by modulating the deposition of H3K4me3.

Genomic distribution was analyzed using the "genomation" R package. GO pathway analysis was performed using the "clusterProfiler" R package. The sequencing information of CUT&Tag data used in this study is summarized in Supplementary Table S1.

Clustering analysis

Promoters were defined as ±2 kb regions flanking the annotated TSS. Reads in promoters were counted using the "coverage" command in bedtools (v2.26.0) and further normalized to RPKM. The k-means clustering of H3K4me3 and H3K27me3 enrichment at promoters was conducted using the "kmeans" function in R.

RNA-seq and data analysis

Total RNA was extracted from sorted CD4⁺YFP⁺ T_{reg} cells using the RNeasy Plus Mini Kit (Qiagen, #74134), following the manufacturer's protocol. RNA-Seq libraries were constructed and sequenced by Haplox (Nanchang, China), using an Illumina platform with paired-end reads of 150 bp. RNA-seq data was obtained from GSE82076. Raw reads were trimmed to 50 bp and mapped to the mouse genome (mm9) using TopHat (v2.1.1) with default parameters. Only uniquely mapped reads were kept for downstream analysis. The RNA abundance of each gene was quantified using Cufflinks (v2.2.1).

Single-Cell RNA-Sequencing

A total of 300,000 sort-purified CD4⁺YFP⁺ T_{reg} cells from *Foxp3*^{cre} and *Foxp3*^{cre} *Cxhc1*^{fl/fl} were resuspended in BD Pharmingen Stain Buffer (FBS) (554656; BD). Single cells were isolated using a chromium controller (BD platform, BD Bioscience) according to the manufacturer's instructions, as previously described²⁷. The single cells were labeled with sample tags using the BD Mouse Immune Single-Cell Multiplexing Kit (633793; BD). Following standard protocols, cDNA amplification and library construction were performed to generate scRNA-seq libraries.

Targeted scRNA-seq data processing

The raw FASTQ files were processed by BD Rhapsody using the Targeted analysis pipeline. After alignment and filtering, the distribution-based error correction (DBEC)-adjusted molecules were loaded into R studio (version 4.3.2). All subsequent analyses were performed using the package Seurat (version 4.4.0) with default parameters. Specifically, the scRNA-seq data counts were log-normalized. All targeted genes were scaled and then were used for Principal components analysis (PCA). The batch effects were removed by the HarmonyMatrix function in the Harmony package (version 1.1.0). The first 20 principal components were used to calculate nonlinear dimensionality reduction using RunUMAP. Differential gene expression (DEGs) between clusters was assessed using the FindAllMarkers function. The clusters were then annotated based on DEGs. Barplots were generated using ggplot2 (version 3.4.4). Heatmaps were generated using pheatmap (version 1.0.12).

Analysis of the single-cell TCR-seq repertoire

Raw V(D)J fastq reads were processed using BD Rhapsody Pipeline and then were analyzed using scRepertoire (version 1.12.0). The TCR clonotype was called using the nucleotide sequence of the CDR3 region for both TCR alpha and beta chains. For cells with multiple chains, the top two clonotypes with the highest expression were selected for downstream analysis. Clonal overlap between different cell types was calculated using the clonalOverlap function of scRepertoire. A clonotype was defined as expansion if it could be detected in at least two cells.

scRNA-seq trajectory analysis

UMAP embeddings obtained from the Seurat package were projected into the Slingshot (version 2.10.0) package to construct pseudotime Trajectories for T_{reg} cells. Naïve subsets were set as the root state.

Whole genome bisulfite sequencing (WGBS) and data analysis

Sorted CD4⁺YFP⁺ T_{reg} cells (3 × 10⁶) were lysed in cell lysis buffer to release DNA. The bisulfite-treated DNA was used to prepare the sequencing library. DNA libraries were transferred to the Illumina Platform for sequencing using 150 bp paired-end reads. Raw reads were trimmed using TrimGalore (v0.4.4) with default parameters. Subsequently, the reads were mapped against the mm9 reference genome using Bismark v0.19.0 with parameters '-bowtie2'. PCR duplicates were removed and the methylation levels were calculated using 'bismark_methylation_extractor'. We calculated the mean CpG methylation levels of various genome elements: promoter, 5'-UTR, exon, intron, 3'-UTR, genebody, intergenic, CGIs, and repeats using in-house scripts. The sequencing information of WGBS data used in this study is summarized in Supplementary Table S2.

Statistical analysis

The statistical significance analysis was performed using Prism 8.0 (GraphPad). Error bars are presented as mean ± SD. *P* values of < 0.05 were deemed statistically significant (**P* < 0.05, ***P* < 0.01, ****P* < 0.001 and *****P* < 0.000). Statistical analyses were performed with unpaired Student's *t*-test or two-way ANOVA and Holm-Sidak post hoc test.

Results

FOXP3 binds regulatory loci primed for activation and repression in T_{reg} cells

FOXP3-mediated gene expression is well-recognized, with several studies highlighting its dual role as both a transcriptional activator and repressor^{28–32}. However, the regulation of T_{reg}-specific gene transcription by FOXP3 through intermediary mechanisms, such as epigenetic factors, remains incompletely understood. To address this, we employed CUT&Tag to generate global H3K4me3 maps in T_{reg} cells and compared these with an H3K27me3 ChIP-Seq dataset from Wei *et al.*³³ at previously identified FOXP3-bound loci (Konopacki *et al.* 2019)³⁴. This comparison aimed to identify FOXP3-dependent genes associated with H3K4me3 and H3K27me3 deposition. As expected, H3K4me3 was enriched at gene promoters (fig. S1, A and B). A Venn diagram revealed a significant overlap between FOXP3 binding sites and H3K4me3 peaks, with minimal overlap with H3K27me3 peaks (Fig. 1, A and B). The overlapping regions between FOXP3 binding sites and H3K4me3 or H3K27me3 peaks were predominantly located at promoters (Fig. 1, A and B). We then clustered the promoters into four groups based on the enrichment of H3K4me3 and H3K27me3. Clusters 1 and 3 showed strong enrichment of H3K4me3; cluster 2 was enriched with H3K27me3; and cluster 4 showed weak enrichment of both modifications. FOXP3 enrichment analysis revealed that FOXP3 preferentially binds to the promoters of clusters 1 and 3, which display high levels of H3K4me3 (Fig. 1C). Correspondingly, genes in these clusters exhibited high transcription levels, as demonstrated by our reanalysis of previously published RNA-seq data (Oh *et al.* 2017) shown in the right panels of Figure 1D³⁵. In contrast, genes with H3K27me3 enrichment at their promoters were transcribed at low levels. Gene Ontology (GO) analysis of these four clusters revealed that cluster 1 was significantly enriched in genes involved in mRNA processing, covalent chromatin modification, and histone modification, while cluster 3 was enriched in genes related to DNA repair and mitochondrion organization (Fig. 1E). Cluster 2, on the other hand, was enriched in the pattern specification process, whereas cluster 4 showed no significant correlation with T_{reg} cells. As expected, signature T_{reg} cell genes such as *Tnfrsf18*, *Nrp1*, *Stat5a*, *Lag3*, *Icos*, and *Pdcd1* were enriched in clusters 1 and 3, which exhibited strong H3K4me3 marks (fig. S1C). Conversely, FOXP3-bound sites in cluster 2 were specifically enriched for H3K27me3, including genes like *Hic1*, *Trp73*, and *Rnf157*, which are associated with inflammatory responses (fig. S1C). These observations support the hypothesis that FOXP3 primarily functions as an activator by promoting H3K4me3 deposition in T_{reg} cells.

CXXC1 interacts with FOXP3 and binds H3K4me3-enriched sites in T_{reg} cells

We conducted an enrichment analysis of known motifs at the overlapping peaks of FOXP3 ChIP-seq and H3K4me3 CUT&Tag in T_{reg} cells to identify epigenetic factors that directly interact with FOXP3 to mediate chromatin remodeling and transcriptional reprogramming. Motif analysis of the overlapping peaks between FOXP3 binding sites and regions enriched in H3K4me3 revealed that, in addition to transcription factors, the most abundant motif associated with H3K4me3 was the epigenetic factor CXXC1 (fig. S2A). To investigate this further, we performed CUT&Tag for endogenous CXXC1 in T_{reg} cells to examine the genome-wide co-occupancy of CXXC1 and FOXP3. Over half of these CXXC1 binding sites were located at promoter regions (fig. S2B). Additionally, CXXC1 exhibited strong binding at transcription start site (TSS) and CpG islands (CGIs) (Fig. 2A and fig. S2C). As illustrated by the Venn diagram (Fig. 2B), more than half of the FOXP3-bound genes and H3K4me3-enriched genes were also bound by CXXC1. Similarly, more than half of CXXC1 peaks were overlapped with FOXP3 peaks (fig. S2D). Furthermore, the CXXC1-specific and

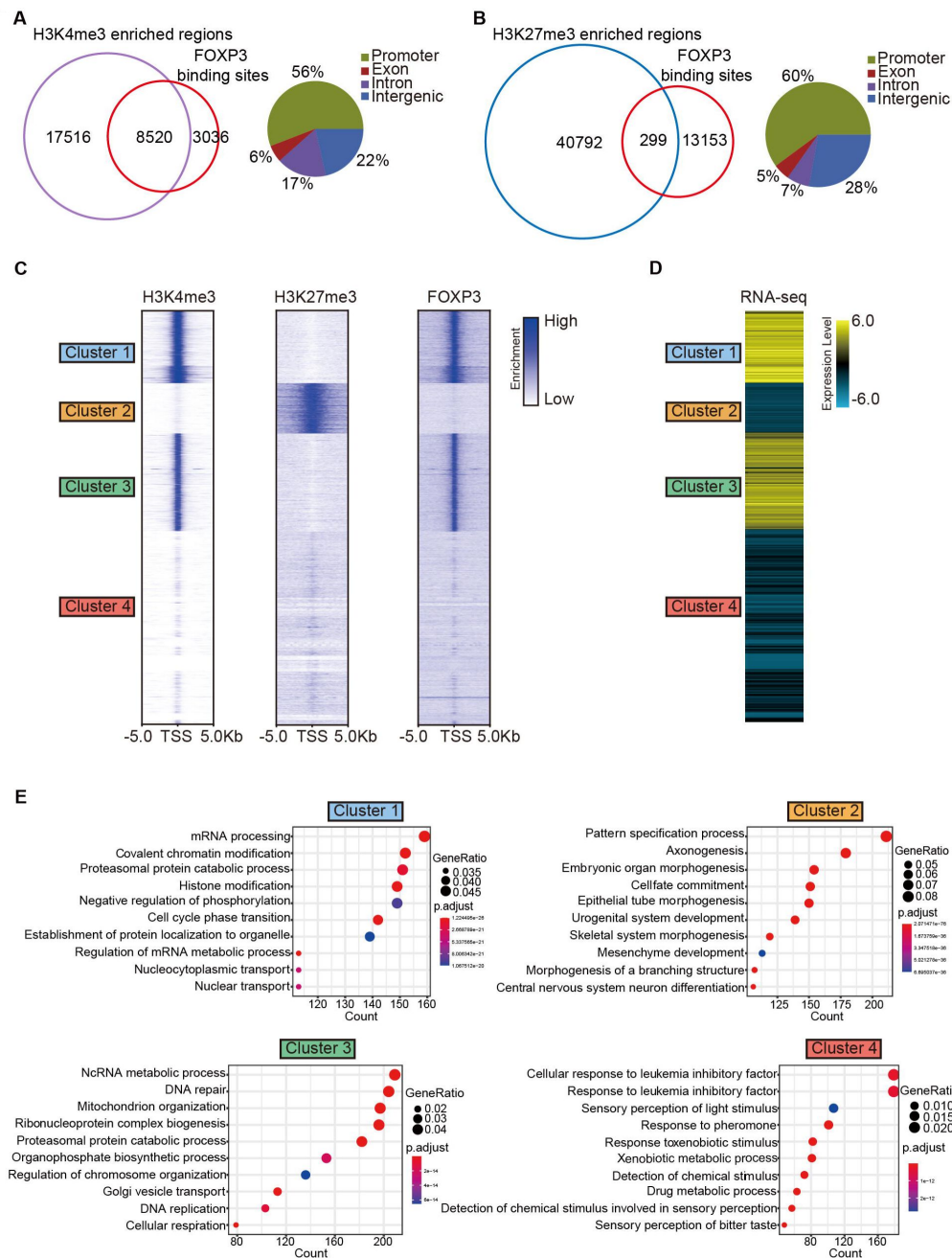


Fig. 1.

H3K4me3 is required for FOXP3-dependent gene activation in T_{reg} cells.

(A) Venn diagram showing overlap of H3K4me3 enriched regions (this study) and FOXP3 binding sites (Konopacki *et al.* 2019) in sorted $CD4^{+}YFP^{+} T_{reg}$ cells (left). Genomic distribution of overlapped peaks (right). Note that the overlapped peaks are predominantly enriched at promoters.

(B) Venn diagram showing overlap of H3K27me3 enriched regions (Wei *et al.* 2009) and FOXP3 binding sites in T_{reg} cells (left). Genomic distribution of overlapped peaks (right). Note that the overlapped peaks are predominantly enriched at promoters.

(C) Heatmap showing enrichment of H3K27me3, H3K4me3, and FOXP3 surrounding the TSS. Unsupervised K-means clustering was conducted on H3K27me3 and H3K4me3 signals.

(D) Heatmap showing gene expression levels in T_{reg} cells (RNA-seq data was obtained from Oh *et al.* 2017). The clusters were consistent as in Fig. 1C.

(E) Gene Ontology (GO) pathway analysis of different clusters.

FOXP3-specific binding sites also demonstrated modest binding of FOXP3 and CXXC1, respectively (Fig. 2C). These findings indicate that FOXP3 and CXXC1 share a substantial number of target genes in T_{reg} cells. To confirm this interaction, we further validated the reciprocal immunoprecipitation of both endogenous and exogenous CXXC1 and FOXP3 (Fig. 2D and fig. S2E). An immunofluorescence assay revealed predominant colocalization of CXXC1 with FOXP3 in the nucleus (Fig. 2E). Overall, these results suggest that CXXC1 primarily functions as a coactivator of FOXP3-driven transcription in T_{reg} cells.

Complete ablation of *Cxxc1* in T_{reg} cells

leads to a fatal autoimmune disease

To investigate the role of CXXC1 in T_{reg}-cell homeostasis and function, we generated *Foxp3*^{Cre}*Cxxc1*^{fl/fl} mice (conditional knockout [cKO] mice) by crossing *Cxxc1*^{fl/fl} with *Foxp3*^{YFP-Cre}³⁶ mice, thereby specifically deleting *Cxxc1* in T_{reg} cells. The effective depletion of *Cxxc1* in T_{reg} cells was confirmed through quantitative PCR (qPCR) and Western blotting (fig. S3A). Notably, cKO mice appeared normal at birth but later exhibited spontaneous mortality starting around three weeks of age (Fig. 3A). Deletion of *Cxxc1* in T_{reg} cells led to the development of severe inflammatory disease, characterized by reduced body size, stooped posture, crusting of the eyelids, ears, and tail, and skin ulceration, particularly on the head and upper back (Fig. 3, B and C). Additionally, cKO mice developed extensive splenomegaly and lymphadenopathy (Fig. 3D). Histopathological analysis revealed massive lymphocyte and myeloid cell infiltration in the skin, lungs, liver sinusoids, and colon mucosa (Fig. 3E). Moreover, cKO mice displayed a significant increase in CD8⁺ T cell percentages (fig. S3B), along with a marked rise in cells exhibiting an effector/memory phenotype (CD44^{hi} CD62L^{lo}) (Fig. 3F). Furthermore, T cells from cKO mice produced elevated levels of IFN-γ, IL-17, and IL-4 in CD4⁺ T cells, as well as increased IFN-γ production in CD8⁺ T cells (Fig. 3G and fig. S3, C and D). These phenotypes closely resembled those observed in *Foxp3*-deficient mice³⁷ or mice with depleted T_{reg} cells³⁸, suggesting a deficiency in immune suppression.

CXXC1 is necessary for the maintenance of T_{reg}-cell suppressive activity

Despite the development of severe autoimmune disease, we observed an increase in both the absolute number and percentage of FOXP3⁺ T_{reg} cells in the lymph nodes (fig. S4A). The expression level of the FOXP3 protein was only slightly altered in *Cxxc1*-deficient T_{reg} cells (fig. S4B). In an *in vitro* suppression assay, T_{reg} cells from *Foxp3*^{Cre}*Cxxc1*^{fl/fl} and WT mice exhibited similar suppressive effects on naïve T (T_n) cell proliferation (fig. S4C). The expression of the hallmark T_{reg}-cell marker CTLA-4 showed a modest increase in *Cxxc1*-deficient T_{reg} cells compared to WT T_{reg} cells, while the expression of GITR remained unchanged (fig. S4D). To further assess the suppressive capacity of *Cxxc1*-deficient T_{reg} cells *in vivo*, we employed the experimental autoimmune encephalomyelitis (EAE) model. Naïve CD4⁺ T cells from 2D2 mice were co-transferred with T_{reg} cells from either *Foxp3*^{Cre} or *Foxp3*^{Cre} *Cxxc1*^{fl/fl} mice into *Rag1*^{-/-} recipients, and EAE was induced in these recipient mice. Mice that received only naïve CD4⁺ T cells from 2D2 mice developed more severe EAE symptoms (Fig. 4A). The addition of WT T_{reg} cells from *Foxp3*^{Cre} mice slightly mitigated EAE progression and reduced Th17 cells in the spinal cord (Fig. 4, A to D). In contrast, *Foxp3*^{Cre} *Cxxc1*^{fl/fl} T_{reg} cells failed to suppress EAE (Fig. 4, A to D), and the cKO mice showed a significant reduction in T_{reg} cells frequency in CNS tissues (Fig. 4E). Finally, we examined the role of CXXC1 in T_{reg}-cell-mediated suppression using T cell transfer-induced colitis, in which naïve T cells were transferred to *Rag1*^{-/-} recipients either alone or together with WT or *Foxp3*^{Cre} *Cxxc1*^{fl/fl} T_{reg} cells. The transfer of naïve T cells led to weight loss and intestinal pathology in recipient mice (Fig. 4, F and G). Mice receiving WT T_{reg} cells

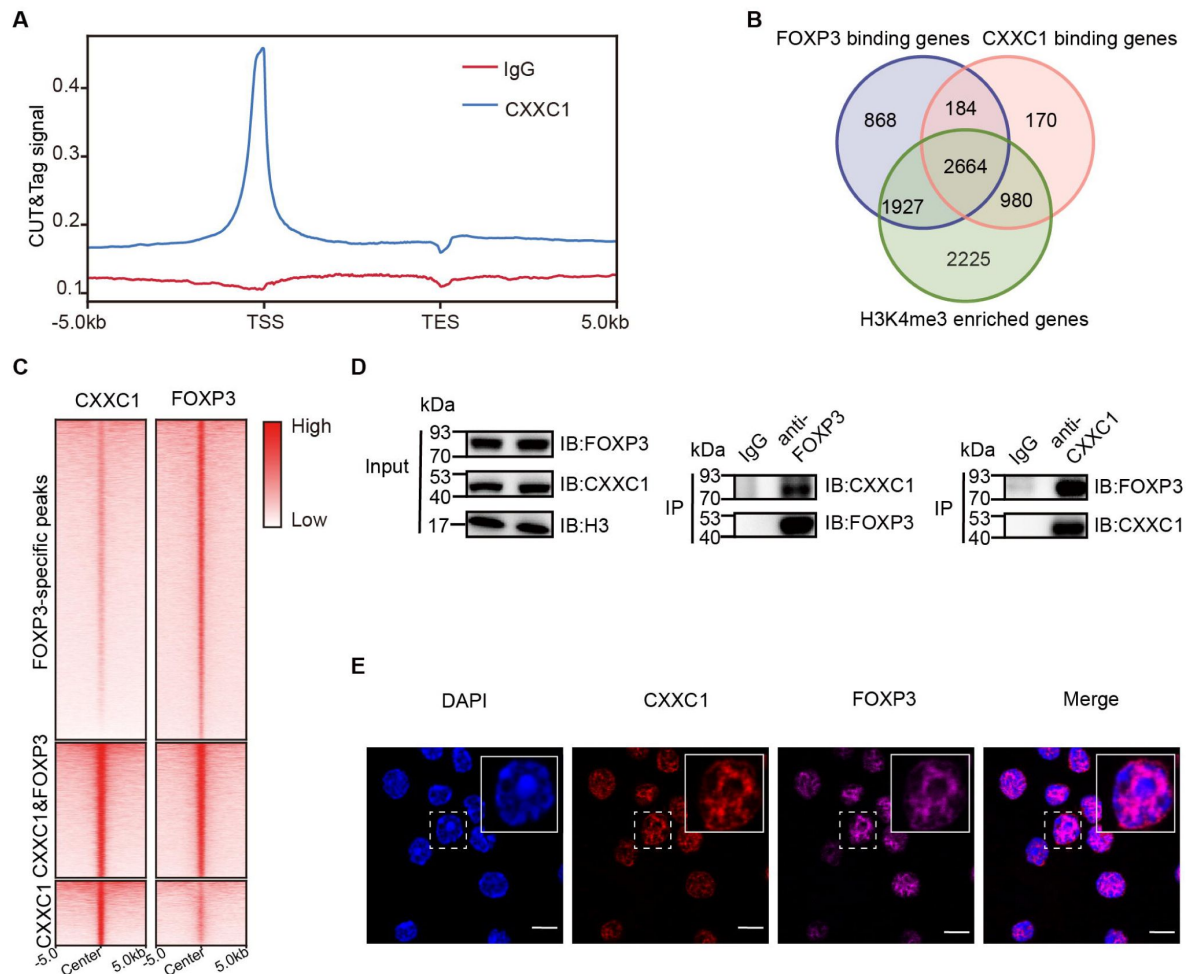


Fig. 2.

CXXC1 interacts with FOXP3 in T_{reg} cell.

- (A) Average CXXC1 CUT&Tag signals around promoters in T_{reg} cells. IgG was used as the control.
- (B) Venn diagrams showing the overlap of FOXP3 binding genes, CXXC1 binding genes, and H3K4me3 enriched genes in T_{reg} cells. Gene promoters covered by FOXP3 binding sites, CXXC1 binding sites, or exhibited high H3K4me3 levels were defined as FOXP3-bound genes, CXXC1-bound genes, or H3K4me3-enriched genes.
- (C) Heatmaps showing FOXP3 ChIP-seq and CXXC1 CUT&Tag signals at indicated regions.
- (D) Interaction between FOXP3 and CXXC1 was assessed by co-IP (forward and reverse) using T_{reg} cell lysates.
- (E) Immunofluorescence for FOXP3 and CXXC1 colocalization in T_{reg} cells. Scale bars, 2 μ m.

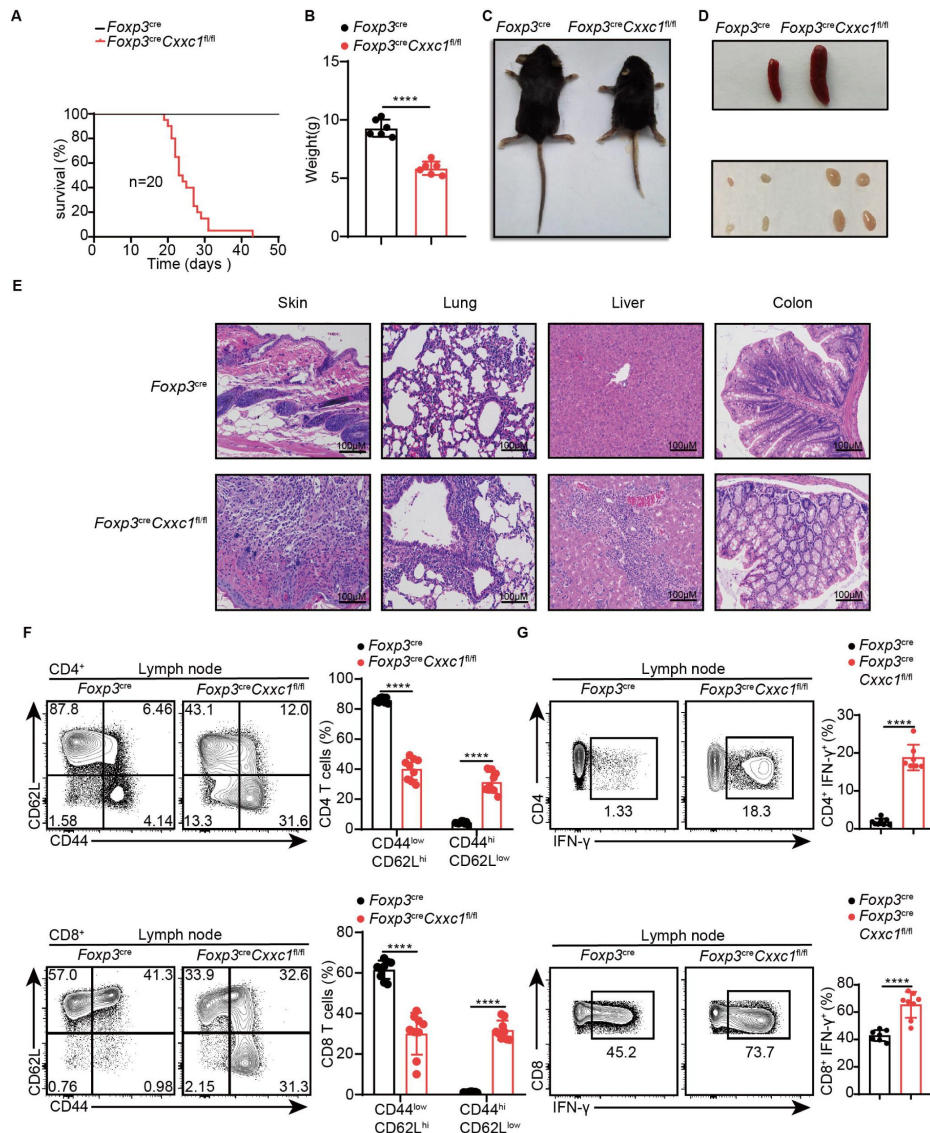


Fig. 3.

Foxp3^{cre}Cxhc1^{fl/fl} mice spontaneously develop a fatal early-onset inflammatory disorder.

(A) Survival curves of *Foxp3^{cre}* (black line) and *Foxp3^{cre}Cxhc1^{fl/fl}* (red line) mice ($n = 20$).

(B) Gross body weight of *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice ($n = 6$).

(C) A representative image of *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice.

(D) Representative images showing the spleen and peripheral lymph nodes from *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice.

(E) Hematoxylin and eosin staining of the skin, lung, liver, and colon from *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice (scale bar, 100 μ m).

(F) Flow cytometry analysis of CD62L and CD44 expression on peripheral lymph node CD4⁺ and CD8⁺ T cells from *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice (left). Right, frequency of CD44^{low}CD62L^{hi} and CD44^{hi}CD62L^{low} population in CD4⁺ or CD8⁺ T cells ($n = 9$).

(G) Lymph node cells from *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice were stimulated ex vivo with PMA + ionomycin for 4h and analyzed for IFN- γ expressing in CD4⁺ or CD8⁺ T cells using flow cytometry (left). Right, percentages of IFN- γ ⁺CD4⁺ or IFN- γ ⁺CD8⁺ T cells in the lymph nodes of *Foxp3^{cre}* and *Foxp3^{cre}Cxhc1^{fl/fl}* mice ($n = 7$).

All mice analyzed were 18-20 days old unless otherwise specified. Error bars show mean $\pm$ SD. The log-rank survival curve was used for survival analysis in A, and multiple unpaired *t*-test or two-tailed Student's *t*-test were used for statistical analyses in B, F, and G (**** $P < 0.0001$). The flow cytometry results are representative of three independent experiments.

continued to gain weight (**Fig. 4F**), whereas those that received T_{reg} cells from cKO mice were unable to prevent colitis and exhibited a reduced percentage of T_{reg} cells (**Fig. 4, F** to H). These findings underscore the critical role of CXXC1 in maintaining T_{reg} cell function *in vivo*.

T_{reg} cell lineage stability and proliferation depend upon CXXC1

T_{reg} cells harbor a diverse T-cell receptor (TCR) repertoire, which likely play a critical role in their immune suppression function^{39–41}. To explore the role of CXXC1 in T_{reg} -mediated suppression, we performed single-cell RNA sequencing (scRNA-seq) combined with TCR sequencing (TCR-seq) on $CD4^+YFP^+$ T_{reg} cells isolated from mouse lymph nodes. After quality control and removal of doublets, 18,577 cells were retained for further analysis. Through unsupervised clustering and uniform manifold approximation and projection (UMAP) analysis, we identified eight distinct T_{reg} cell clusters based on the expression of well-characterized markers, with a particular focus on two clusters of activated T_{reg} cells that exhibited higher expression of markers and gene sets relative to naïve T_{reg} cells (**Fig. 5A** and fig. S5, A to C). A comparison between *Cxxc1*-deficient and WT T_{reg} cells within each cluster revealed a reduction in *Cxxc1*-deficient cells in the naïve subsets, while an increase was observed in the $Gzmb^+$ and $H2-Eb1^+$ subsets (**Fig. 5B**). To further elucidate the transition of T_{reg} cells along a dynamic biological timeline, we constructed pseudo-time trajectories using Slingshot⁴². The pseudo-time gradient depicted a progression from quiescent to activated T_{reg} cells, ultimately encompassing the $Gzmb^+$ and $H2-Eb1^+$ subsets (**Fig. 5C**). Given the antigen-specific suppression capabilities of T_{reg} cells^{43,44}, we examined their clonal expansion. The analysis revealed that expanded WT TCR clonotypes ($n \geq 2$) were predominantly distributed among the $Nt5e^+$ subsets, while *Cxxc1*-deficient T_{reg} cells showed expanded clonotypes primarily within the $Gzmb^+$ and $H2-Eb1^+$ subsets (**Fig. 5D** and Fig. S5D). TCR sharing analysis indicated significant clonotype sharing among various clusters of WT T_{reg} cells, suggesting a degree of homogeneity. However, the reduced TCR sharing in *Cxxc1*-deficient cells implies that decreased TCR diversity may impair the suppressive activity of T_{reg} cells (**Fig. 5E**)⁴¹. Furthermore, the *Cxxc1*-deficient group exhibited lower expression of several T_{reg} -specific genes associated with suppressive functions, such as *Nt5e*, *Il10*, *Pdcd1*, *Klrg1*, as well as genes that inhibit effector T cells differentiation, including *Sell* and *Tcf7* (**Fig. 5F** and fig. S5E). Conversely, *Cxxc1*-deficient T_{reg} cells demonstrated elevated expression of *Gzmb*, *Il2ra*, and *Cd69* compared to WT, reflecting a profile indicative of increased activation (**Fig. 5, F** and G). Additionally, we observed increased expression of genes linked to Th1-type inflammation, such as *Ifng*, *Tbx21*, and *Hif1a*, in *Cxxc1*-deficient T_{reg} cells, likely due to extreme inflammatory conditions (**Fig. 5, F** to H). The proportion of $FOXP3^+Ki67^+$ T_{reg} cells was lower in cKO mice compared to WT mice (**Fig. 5I**). These findings underscore the crucial role of CXXC1 in maintaining T_{reg} cell stability.

Cxxc1 deletion leads to a reduction in essential T_{reg} cell markers

To confirm that the deficiency in T_{reg} cell function in T_{reg} -specific *Cxxc1*-deficient animals is due to intrinsic defects caused by *Cxxc1* deficiency, rather than severe autoimmune inflammation in *Foxp3^{Cre} Cxxc1^{fl/fl}* mice, we examined *Cxxc1*-sufficient and *Cxxc1*-deficient T_{reg} cell subsets in heterozygous *Foxp3^{Cre/+} Cxxc1^{fl/fl}* (designated as “het-KO”) and littermate *Foxp3^{Cre/+} Cxxc1^{fl/+}* (designated as “het-WT”) female mice (**Fig. 6A**). Notably, het-KO female mice did not exhibit overt signs of autoimmunity, as random X-chromosome inactivation led to the coexistence of both *Cxxc1*-cKO and *Cxxc1*-WT T_{reg} cells. However, both the frequency and absolute numbers of $FOXP3^+YFP^+$ T_{reg} cells within the total T_{reg} population were reduced in het-KO mice compared to their counterparts in het-WT littermates, indicating that *Cxxc1* deficiency imposes a competitive disadvantage on T_{reg} cells (**Fig. 6B**). Additionally, *Cxxc1*-deficient YFP^+ T_{reg} cells failed to upregulate the proliferation marker Ki-67 (**Fig. 6C**). Moreover, YFP^+ T_{reg} cells in het-KO female mice showed reduced expression of key genes essential for suppressive function, including *Icos*, *Il2ra*, *Ctla4*, and *Tnfrsf18*, compared to YFP^+T_{reg} cells from the same mice (**Fig. 6D**). To

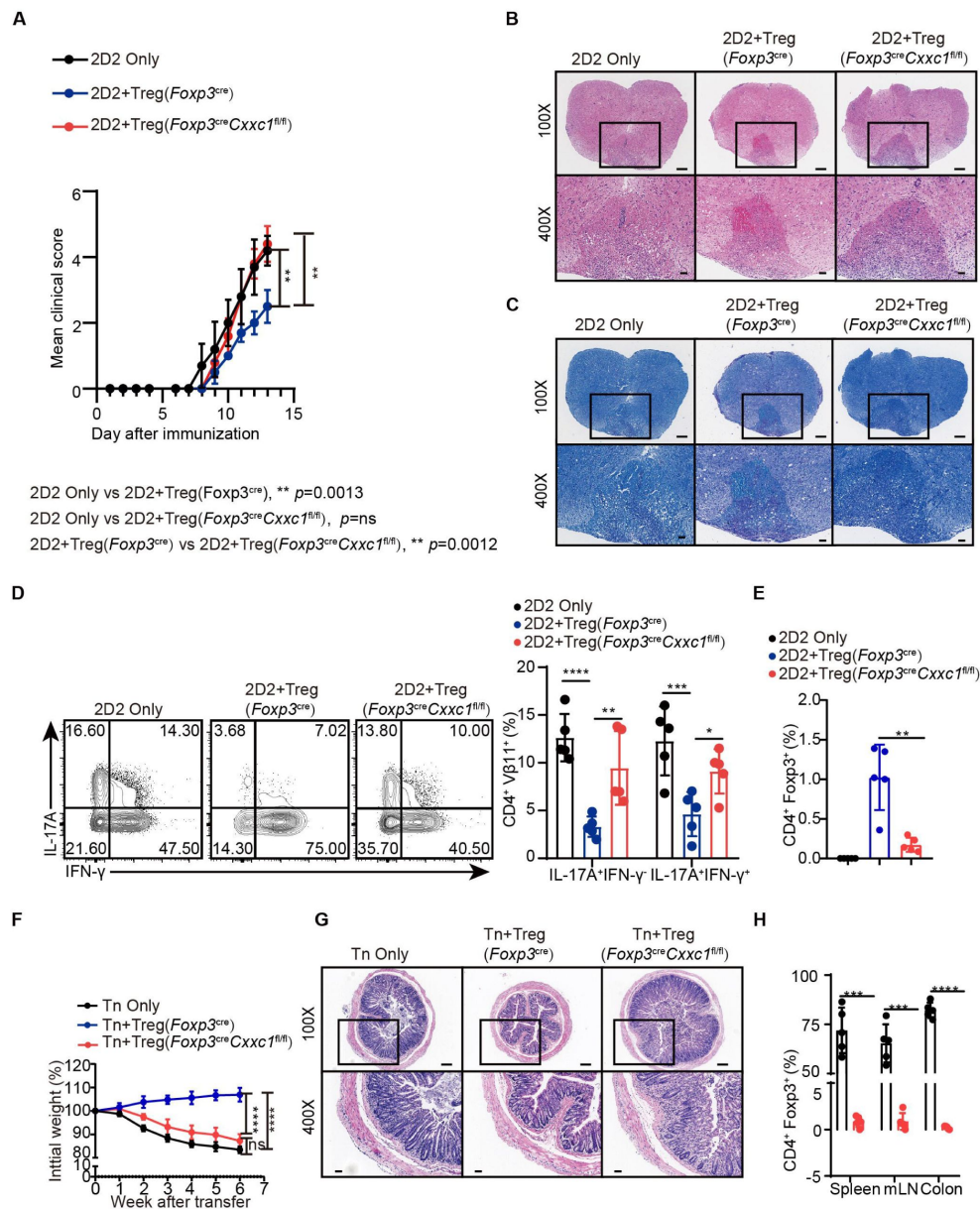


Fig. 4.

CXXC1 is essential for T_{reg} cells to suppress T cell-mediated EAE and colitis.

(A) Mean clinical scores for EAE in *Rag1*^{-/-} recipients of 2D2 CD4⁺ T cells, either alone or in combination with *Foxp3*^{cre} or *Foxp3*^{cre}*Cxxc1*^{fl/fl} mice after immunization with MOG35–55, complete Freund's adjuvant (CFA), and pertussis toxin (n=5). (B and C) Representative histology of the spinal cord of *Rag1*^{-/-} mice after EAE induction. Hematoxylin and eosin (H&E) staining (upper), Luxol fast blue (F&B) staining (lower). Scale bars, 50 μm (400×) and 200 μm (100×). (D) Representative flow cytometry plots and quantification of the percentages of IFNγ⁺ or IL-17A⁺ CD4⁺Vβ11⁺ T cells (n=5). (E) Statistical analysis of the percentage CD4⁺ FOXP3⁺ T_{reg} cell in CNS tissues 14 days after EAE induction. (F) Changes in body weight of *Rag1*^{-/-} mice after colitis induction (n=6). (G) Haematoxylin and eosin (H&E) staining of colons from T cell-induced colitis mice 6 weeks after T cell transfer. Scale bars, 50 μm (400×) and 200 μm (100×). (H) Statistical analysis of the percentage CD4⁺ FOXP3⁺ T_{reg} cell in the spleen, mesenteric lymph nodes, and colon 6 weeks after colitis induction.

Error bars show mean ± SD. *P* values are determined by a two-tailed Student's *t*-test or two-way ANOVA and Holm–Sidak post hoc test (A, D, and E). (* $P<0.05$, ** $P<0.01$, *** $P<0.001$, **** $P<0.0001$).

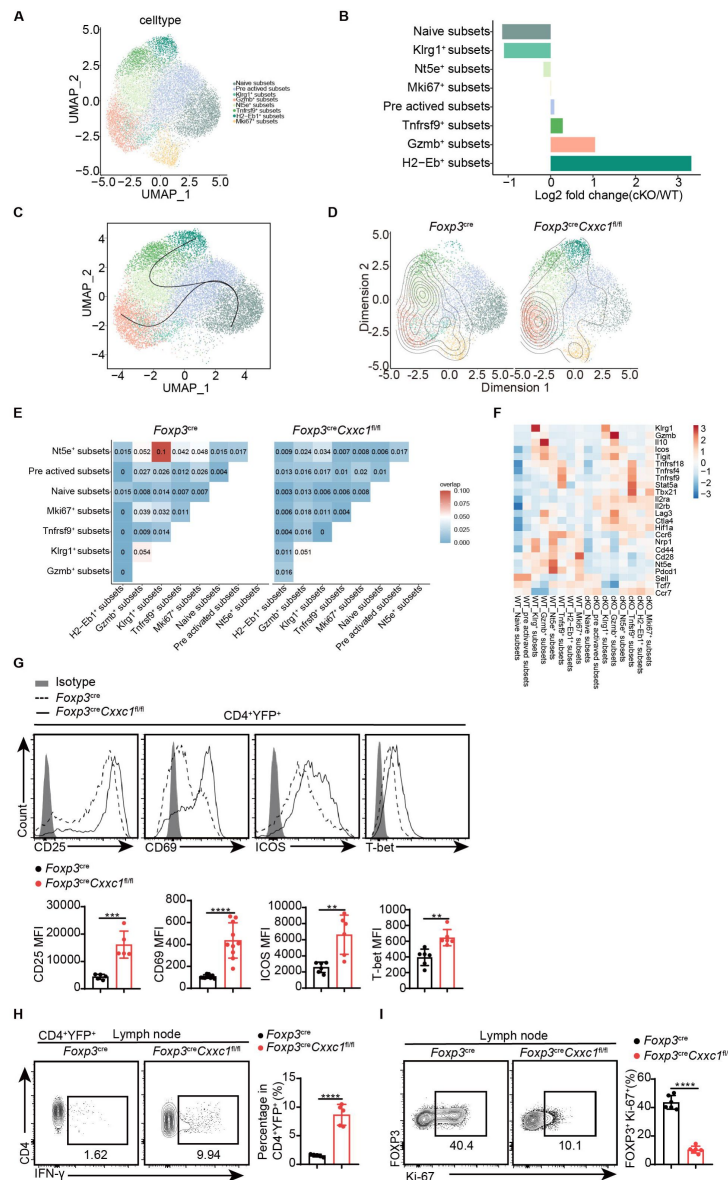


Fig. 5.

Single-cell transcriptomics reveals distinct T_{reg} cell populations.

(A) UMAP plot showing clusters identified based on variable gene expression of sorted CD4⁺YFP⁺ T_{reg} cells. Each dot represents a cell, and each color corresponds to a different population of cell types. Clustering analysis revealed 8 distinct T_{reg} cell populations.

(B) Mean fold changes in cluster abundance between *Foxp3*^{cre} and *Foxp3*^{cre}*Cxxc1*^{fl/fl} mice.

(C) Pseudotime trajectories of T_{reg} cells based on Slingshot, color-coded by T_{reg} cell subpopulations.

(D) Visualization of density and clonotype richness across T_{reg} clusters from *Foxp3*^{cre} and *Foxp3*^{cre}*Cxxc1*^{fl/fl} mice.

(E) TCR sharing of expanded clonotypes across all possible combinations of T_{reg} cells from *Foxp3*^{cre} and *Foxp3*^{cre}*Cxxc1*^{fl/fl} mice.

(F) Heatmap showing Z scores for the average expression of T_{reg}-specific genes in each cluster between *Foxp3*^{cre} and *Foxp3*^{cre}*Cxxc1*^{fl/fl} mice.

(G-H) Representative flow cytometry plots and quantification of (G) expression of CD25, CD69, ICOS, T-bet, and (H) IFN-γ in CD4⁺YFP⁺ T_{reg} cells from *Foxp3*^{cre} and *Foxp3*^{cre}*Cxxc1*^{fl/fl} mice (n=5 CD25, n=10 CD69, n=5 ICOS, n=6 T-bet, n=5 IFN-γ).

(I) Ki-67 expression (left) and frequency (right) in CD4⁺FOXP3⁺ T_{reg} cells from *Foxp3*^{cre} and *Foxp3*^{cre}*Cxxc1*^{fl/fl} mice (n=6).

Error bars show mean ± SD. P values are determined by a two-tailed Student's *t*-test (G-H). (***P* < 0.01, ****P* < 0.001, *****P* < 0.0001). The flow cytometry results are representative of three independent experiments.

investigate the molecular program affected by the deletion of *Cxhc1* in T_{reg} cells, we performed RNA-sequencing (RNA-seq) analysis on $CD4^{+}YFP^{+} T_{reg}$ cells isolated from het-WT and het-KO mice. We then conducted a differential gene expression (DGE) analysis based on the RNA-seq data. Among all expressed genes, 865 were upregulated (“Up” genes) and 761 were downregulated (“Down” genes) by ≥ 1.5 -fold in $CD4^{+}YFP^{+} T_{reg}$ cells from het-KO mice compared to het-WT mice, with a false discovery rate (FDR)-adjusted *P*-value cutoff of 0.05 (fig. S6A). Gene Ontology (GO) enrichment analysis revealed that the downregulated genes in *Cxhc1*-deficient T_{reg} cells were predominantly enriched in pathways related to the negative regulation of immune system process and regulation of cell-cell adhesion (fig. S6B). The *Cxhc1*-deficient T_{reg} cells also showed reduced expression of several genes associated with T_{reg} cell suppressive function, including *Il10*, *Tigit*, *Lag3*, *Icos*, *Nt5e* (encoding CD73) and *Itgae* (encoding CD103) (fig. S6C). Thus, although YFP^{+} WT T_{reg} cells can prevent autoimmunity in het-KO mice, the absence of *Cxhc1* in $YFP^{+} T_{reg}$ cells leads to the dysregulation of key T_{reg} cell markers under steady-state conditions.

The FOXP3-CXXC1 complex regulates the expression of key factors in T_{reg} cells that are associated with the breadth of H3K4me3

CXXC1 binds to unmethylated CpG DNA via its N-terminal CXXC finger domain, facilitating its interaction with DNA methyltransferase 1 (DNMT1). This binding stabilizes the DNMT1 protein, thereby regulating DNA methylation^{45,46}. To investigate whether CXXC1 depletion affects DNA methylation in T_{reg} cells, we performed whole genome bisulfite sequencing (WGBS) on T_{reg} cells isolated from both WT and cKO mice. On average, *Cxhc1*-deficient T_{reg} cells exhibited no significant changes in DNA methylation at gene loci or across genome-wide CpG sites, irrespective of chromosomal region (fig. S7, A to C). Furthermore, *Cxhc1* knockout T_{reg} cells did not show a significant increase in DNA methylation at key T_{reg} signature gene loci (fig. S7D). Given the pivotal role of MLL4-mediated H3K4me1 in establishing the enhancer landscape and facilitating long-range chromatin interactions during T_{reg} cell development¹⁵, we performed CUT&Tag to assess changes in H3K4me1 levels in *Cxhc1*-deficient T_{reg} cells. This analysis revealed that H3K4me1 levels were similar in both WT and *Cxhc1*-deficient T_{reg} cells (fig. S7, E to G).

While H3K4me3 modifications typically form sharp 1- to 2-kb peaks around promoters, some genes exhibit broader H3K4me3 regions, referred to as broad H3K4me3 domains (H3K4me3-BDs), which can extend to cover part or all of the gene’s coding sequences (up to 20 kb)^{47,48}. Broad H3K4me3 domains are preferentially associated with genes essential for the identity or function of specific cell types^{47,49} and have been implicated in enhancing transcriptional elongation and increasing enhancer activity⁴⁹. To further explore the connection between broad H3K4me3 domains and the expression of immune-regulatory genes, we analyzed genes containing broad H3K4me3 regions. We classified the H3K4me3 domains surrounding transcription start sites (TSSs) into three categories: broad (more than 5 kb), medium (between 1 and 5 kb), and narrow (less than 1 kb) (Fig. 7A). Notably, *Cxhc1*-deficient T_{reg} cells exhibited weaker H3K4me3 signals compared to WT cells within the broad H3K4me3 domains where CXXC1 binding is prominent (Fig. 7, A and B). Using the criteria established by Benayoun *et al.*⁴⁷, which defines the top 5% of the widest H3K4me3 domains as BDs, we observed similar enrichment results (fig. S7, H and I). We then compared three groups of genes: BD-associated genes with reduced H3K4me3 levels following *Cxhc1* deletion, genes with direct CXXC1 binding, and genes with direct FOXP3 binding. The Venn diagram revealed that the majority of genes (283 out of 294, 96%) with CXXC1 binding and reduced H3K4me3 levels overlap with FOXP3-bound genes, suggesting that CXXC1 presence enhances FOXP3 binding within the broad H3K4me3 domain (Fig. 7C). Furthermore, GO term analysis indicated that BD-associated genes are enriched in biological processes related to the negative regulation of immune system processes (Fig. 7D). Genome browser views displayed the enrichments of FOXP3, CXXC1, and H3K4me3 at key signature genes in T_{reg} cells, such as *Ctla4*, *Il2ra*, *Icos*, and *Tnfrsf18*, with lower H3K4me3 densities observed at these loci in *Cxhc1*-deficient T_{reg} cells (Fig. 7E). Similar patterns were observed at core genes involved in T_{reg} homeostasis

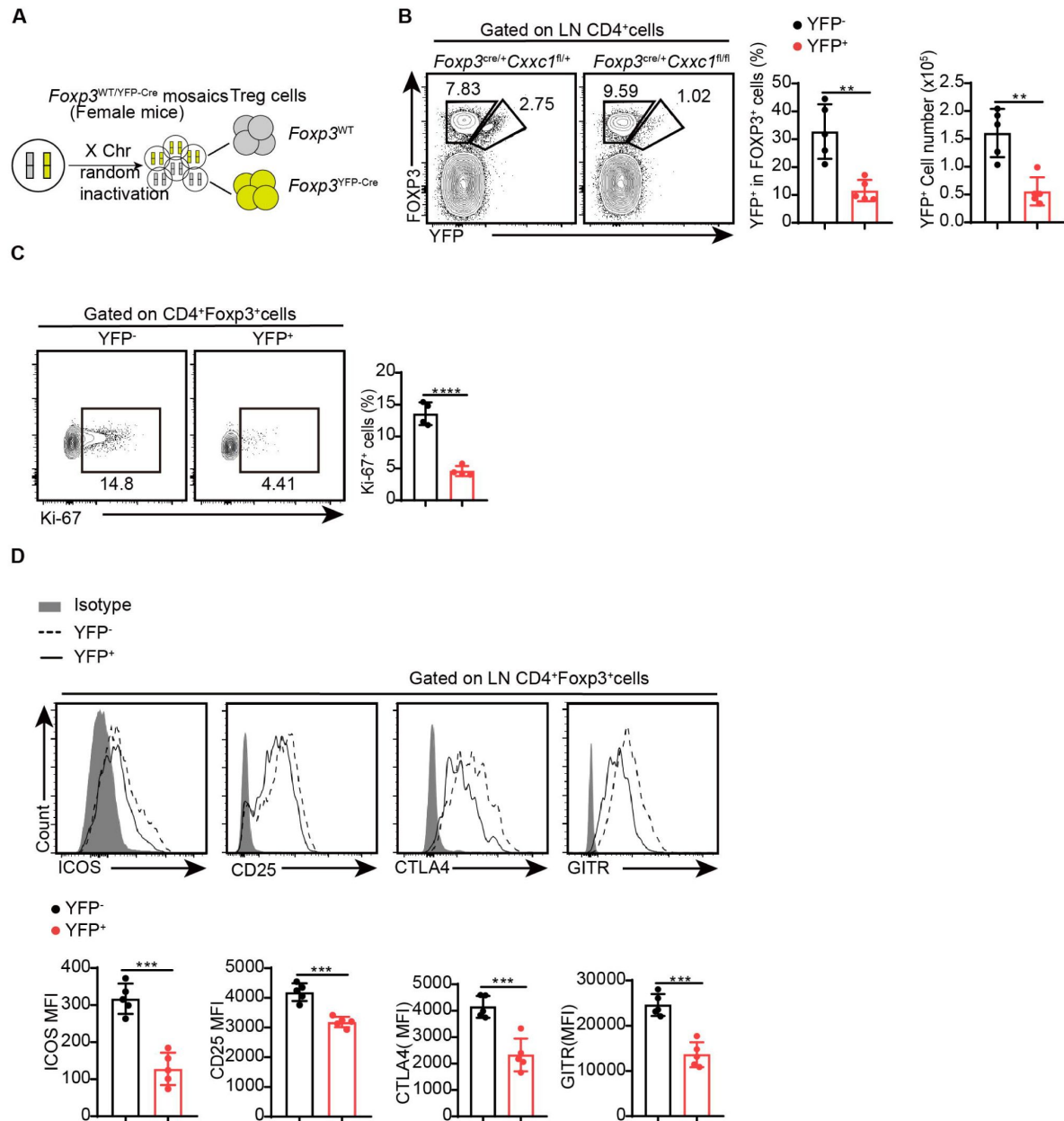


Fig. 6.

$Cxhc1$ -deficient T_{reg} cells in het-KO female mice exhibit reduced suppression markers under steady-state conditions.

(A) Schematic representation of wild-type and Cre-positive T_{reg} cells in female $Foxp3^{Cre/+}$ mice.
 (B) Flow cytometry analysis of the YFP⁺FOXP3⁺ (WT) and YFP⁺FOXP3⁺ (KO) T_{reg} cells in $Foxp3^{Cre/+}$ $Cxhc1^{fl/+}$ (het-WT) and $Foxp3^{Cre/+}$ $Cxhc1^{fl/fl}$ (het-KO) female mice (left), along with the frequency and absolute numbers of YFP⁺ cells within the total T_{reg} population (right) (n=5).
 (C) Flow cytometry analysis of Ki-67 expression (left) and MFI (right) in YFP⁻ and YFP⁺ cells within the CD4⁺FOXP3⁺ T_{reg} cells from 6- to 8-week-old het-KO female mice (n=4).
 (D) Representative flow cytometry plots and quantification of ICOS, CD25, CTLA4, and GITR expression in YFP⁻ and YFP⁺ cells within CD4⁺FOXP3⁺ T_{reg} cells from het-KO female mice (n=5).
 Error bars show mean $\pm$ SD. P values are determined by a two-tailed Student's t -test (G-H). (** P < 0.01, *** P < 0.001, **** P < 0.0001). The flow cytometry results are representative of three independent experiments.

and suppressive function (e.g., *Lag3*, *Nt5e*, *Irf4*, and *Cd28*) (fig. S7, J)^{50,51}. These findings suggest that CXXC1 and FOXP3 collaboratively promote sustained T_{reg} cell homeostasis and function by preserving the H3K4me3 modification at key T_{reg} cell genes.

Discussion

T_{reg} cells specifically express the transcription factor FOXP3, which is essential for maintaining T_{reg} lineage stability and suppressive function⁵². However, FOXP3 alone is insufficient to fully regulate the transcriptional signature and functionality of T_{reg} cells; its interaction with protein partners is crucial for this regulation. In this study, we identify CXXC1 as a key transcription factor that, by acting as a critical cofactor of FOXP3, orchestrates the T_{reg} transcriptional program, which in turn plays a vital role in T_{reg} cell homeostasis and function.

FOXP3 is known to promote histone H3 acetylation at the promoters and enhancers of its target genes, such as *Il2ra*, *Ctla4*, and *Tnfrsf18*, following T_{reg} cell activation, thereby functioning as a transcriptional activator^{53,54}. Conversely, FOXP3 can act as a repressor by silencing target genes like *Il2* and *Ifng* through the induction of histone H3 deacetylation, mediated by the recruitment of histone deacetylases and transcriptional co-repressors^{10,55}. Additionally, FOXP3 exerts this repression by recruiting the Ezh2-containing polycomb repressive complex to target genes during activation, as FOXP3-repressed genes are associated with H3K27me3 deposition and reduced chromatin accessibility^{11,56}. Recent studies have begun to explore the biological significance of H3K4me3 breadth, revealing a positive correlation between H3K4me3 breadth and gene expression^{49,57,58}. These studies also suggest that H3K4me3 breadth contributes to defining specific cell identities during development and disease, including systemic autoimmune diseases like systemic lupus erythematosus and various cancers^{47,59–62}. Our study demonstrates that the loss of *Cxxc1* leads to reduced H3K4me3 levels, predominantly in genes with broader peaks, such as *Il2ra*, *Tnfrsf18*, *Ctla4* and *Icos*, directly impairing their suppressive function in T_{reg} cells. Although our research primarily focused on the role of CXXC1 in T_{reg} cells, it is plausible that similar mechanisms may be operative in other cell types.

The T_{reg}-specific deletion of *Cxxc1* leads to a rapid and fatal autoimmune disorder, characterized by systemic inflammation and tissue damage, underscoring the essential role of CXXC1 in maintaining immune self-tolerance within T_{reg} cells. Interestingly, despite this severe phenotype, our findings show that H3K4me1 levels were comparable between WT and *Cxxc1*-deficient T_{reg} cells. In contrast, MLL4 is critical for T_{reg} cell development in the thymus, primarily by regulating H3K4me1, though it is not required for peripheral T_{reg} cell function¹⁵. Numerous studies investigating the effectors of T_{reg} cell-mediated suppression have identified a wide range of molecules and mechanisms. These include the upregulation of the inhibitory co-stimulatory receptor CTLA-4, which initiates inhibitory signaling; the sequestration of the T-cell growth factor IL-2 via CD25; the secretion of inhibitory cytokines, such as interleukin (IL)-10, IL-35 and TGF-β and the activity of ectoenzymes CD39 and CD73 on the T_{reg} cells surface, which convert extracellular ATP, a potent pro-inflammatory mediator, into its anti-inflammatory counterpart, adenosine^{41,63}. Mechanistically, we demonstrated that CXXC1 interacts with FOXP3 to regulate T_{reg} cell function by trimethylating H3K4 at broad H3K4me3 domains of multiple genes involved in suppressive functions such as *Il2ra*, *Nt5e*, and *Ctla4*. Additionally, MLL1, another KMT, controls T_{reg} cell activation and function by specifically regulating H3K4 trimethylation at genes encoding key T_{reg}-related molecules such as *Tigit*, *Klrg1*, *Tbx21*, *Cxcr3*, and serves as a crucial epigenetic regulator in establishing a stable Th1-T_{reg} lineage¹⁴.

Our results offer provide novel insights into the suppressive functions, heterogeneity, and regulatory mechanisms of T_{reg} cells. Maintaining T_{reg} cell homeostasis and function remains a significant challenge in harnessing T_{reg} cells for the treatment of autoimmune diseases and the prevention of graft rejection.

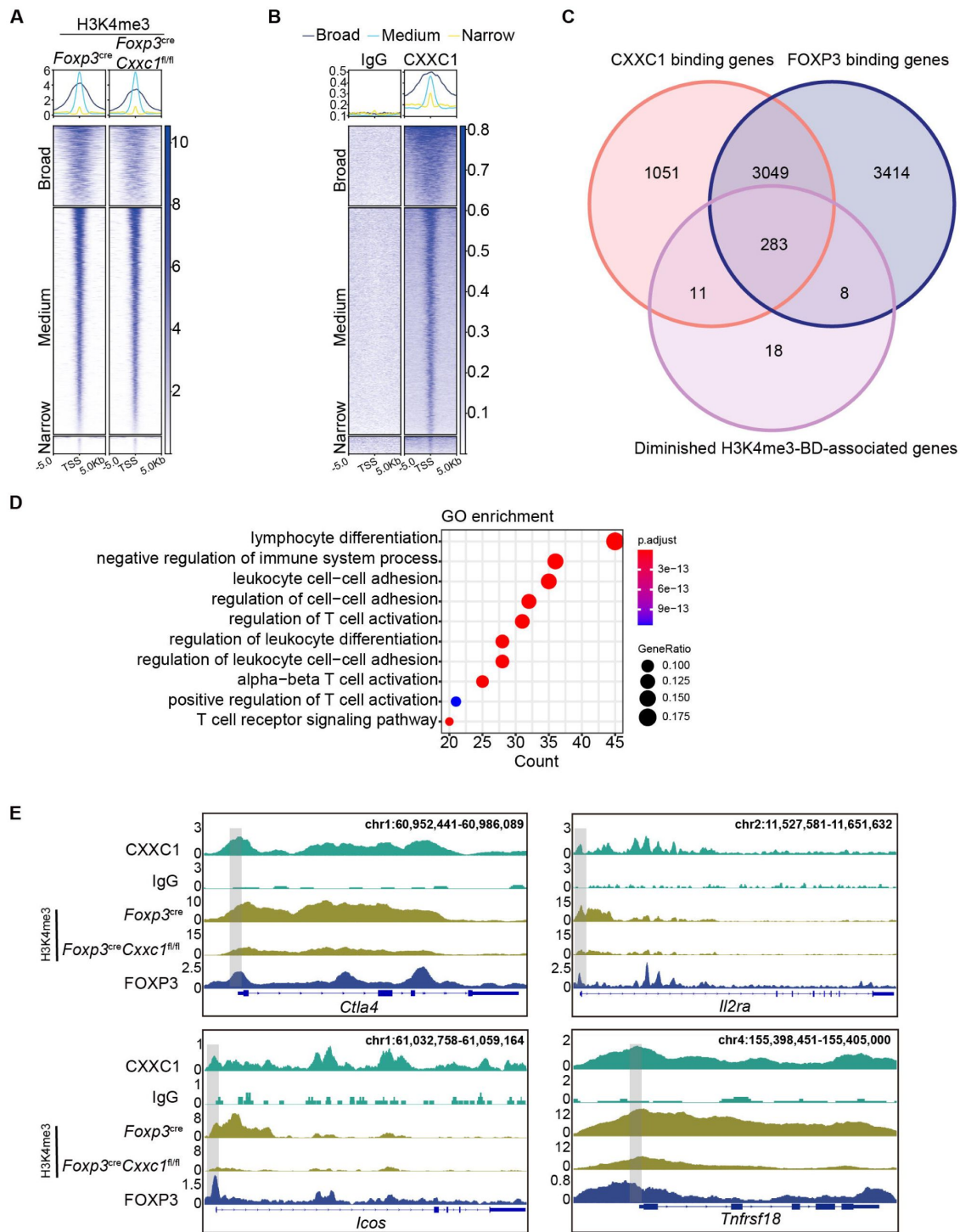


Fig. 7.

CXXC1 regulates *Foxp3*-dependent molecule H3K4 trimethylation in T_{reg} cells.

(A-B) Heatmaps showing H3K4me3(A) and CXXC1(B) signals centered on narrow, medium, and broad domains. The top panel shows the average CUT&Tag signals around indicated domains.

(C) Venn diagrams showing the overlap of FOXP3 binding genes, CXXC1 binding genes, and H3K4me3-BD-associated genes with decreased H3K4me3 levels after CXXC1 depletion in T_{reg} cells.

(D) Gene Ontology (GO) pathway analysis of the overlap peaks.

(E) Representative genome browser view showing the enrichments of FOXP3, CXXC1, and H3K4me3 in T_{reg} cells.

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Additional information

Competing interests

The authors declare that they have no competing interests.

Footnotes

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Author contributions

L.W. and L.S.: supervision, conceptualization, project administration, and writing—review and editing. L.W., L.S. and X.M: funding acquisition. X.M: investigation, methodology, project administration, and writing— original draft. Y.Z. and K.L.: bioinformatics analysis. Y. W., X.L., J.C., C.L., Y.Z., S.W., S.T., Q.X., L.D., X.S., and X.G.: investigation and methodology.

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Reviewer #1 (Public review):

Summary:

This work investigated the role of CXXC-finger protein 1 (CXXC1) in regulatory T cells. CXXC1-bound genomic regions largely overlap with Foxp3-bound regions and regions with H3K4me3 histone modifications in Treg cells. CXXC1 and Foxp3 interact with each other, as shown by co-immunoprecipitation. Mice with Treg-specific CXXC1 knockout (KO) succumb to lymphoproliferative diseases between 3 to 4 weeks of age, similar to Foxp3 KO mice. Although the immune suppression function of CXXC1 KO Treg is comparable to WT Treg in an in vitro assay, these KO Tregs failed to suppress autoimmune diseases such as EAE and colitis in Treg transfer models in vivo. This is partly due to the diminished survival of the KO Tregs after transfer. CXXC1 KO Tregs do not have an altered DNA methylation pattern; instead, they display weakened H3K4me3 modifications within the broad H3K4me3 domains, which contain a set of Treg signature genes. These results suggest that CXXC1 and Foxp3 collaborate

to regulate Treg homeostasis and function by promoting Treg signature gene expression through maintaining H3K4me3 modification.

Strengths:

Epigenetic regulation of Treg cells has been a constantly evolving area of research. The current study revealed CXXC1 as a previously unidentified epigenetic regulator of Tregs. The strong phenotype of the knockout mouse supports the critical role CXXC1 plays in Treg cells. Mechanistically, the link between CXXC1 and the maintenance of broad H3K4me3 domains is also a novel finding.

Weaknesses:

- (1) It is not clear why the authors chose to compare H3K4me3 and H3K27me3 enriched genomic regions. There are other histone modifications associated with transcription activation or repression. Please provide justification.
- (2) It is not clear what separates Clusters 1 and 3 in Figure 1C. It seems they share the same features.
- (3) The claim, "These observations support the hypothesis that FOXP3 primarily functions as an activator by promoting H3K4me3 deposition in Treg cells." (line 344), seems to be a bit of an overstatement. Foxp3 certainly can promote transcription in ways other than promoting H3K4me3 deposition, and it also can repress gene transcription without affecting H3K27me3 deposition. Therefore, it is not justified to claim that promoting H3K4me3 deposition is Foxp3's primary function.
- (4) For the in vitro suppression assay in Figure S4C, and the Treg transfer EAE and colitis experiments in Figure 4, the Tregs should be isolated from Cxhc1 fl/fl x Foxp3 cre/wt female heterozygous mice instead of Cxhc1 fl/fl x Foxp3 cre/cre (or cre/Y) mice. Tregs from the homozygous KO mice are already activated by the lymphoproliferative environment and could have vastly different gene expression patterns and homeostatic features compared to resting Tregs. Therefore, it's not a fair comparison between these activated KO Tregs and resting WT Tregs.
- (5) The manuscript didn't provide a potential mechanism for how CXXC1 strengthens broad H3K4me3-modified genomic regions. The authors should perform Foxp3 ChIP-seq or Cut-n-Taq with WT and Cxhc1 cKO Tregs to determine whether CXXC1 deletion changes Foxp3's binding pattern in Treg cells.

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Reviewer #2 (Public review):

FOXP3 has been known to form diverse complexes with different transcription factors and enzymes responsible for epigenetic modifications, but how extracellular signals timely regulate FOXP3 complex dynamics remains to be fully understood. Histone H3K4 trimethylation (H3K4me3) and CXXC finger protein 1 (CXXC1), which is required to regulate H3K4me3, also remain to be fully investigated in Treg cells. Here, Meng et al. performed a comprehensive analysis of H3K4me3 CUT&Tag assay on Treg cells and a comparison of the dataset with the FOXP3 ChIP-seq dataset revealed that FOXP3 could facilitate the regulation of target genes by promoting H3K4me3 deposition.

Moreover, CXXC1-FOXP3 interaction is required for this regulation. They found that specific knockdown of Cxhc1 in Treg leads to spontaneous severe multi-organ inflammation in mice and that Cxhc1-deficient Treg exhibits enhanced activation and impaired suppression

activity. In addition, they have also found that CXXC1 shares several binding sites with FOXP3 especially on Treg signature gene loci, which are necessary for maintaining homeostasis and identity of Treg cells.

The findings of the current study are pretty intriguing, and it would be great if the authors could fully address the following comments to support these interesting findings.

Major points:

(1) There is insufficient evidence in the first part of the Results to support the conclusion that "FOXP3 functions as an activator by promoting H3K4Me3 deposition in Treg cells". The authors should compare the results for H3K4Me3 in FOXP3-negative conventional T cells to demonstrate that at these promoter loci, FOXP3 promotes H3K4Me3 deposition.

(2) In Figure 3 F&G, the activation status and IFN γ production should be analyzed in Treg cells and Tconv cells separately rather than in total CD4 $^{+}$ T cells. Moreover, are there changes in autoantibodies and IgG and IgE levels in the serum of cKO mice?

(3) Why did Cxxc1-deficient Treg cells not show impaired suppression than WT Treg during in vitro suppression assay, despite the reduced expression of Treg cell suppression assay - associated markers at the transcriptional level demonstrated in both scRNA-seq and bulk RNA-seq?

(4) Is there a disease in which Cxxc1 is expressed at low levels or absent in Treg cells? Is the same immunodeficiency phenotype present in patients as in mice?

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Reviewer #3 (Public review):

In the report entitled "CXXC-finger protein 1 associates with FOXP3 to stabilize homeostasis and suppressive functions of regulatory T cells", the authors demonstrated that Cxxc1-deletion in Treg cells leads to the development of severe inflammatory disease with impaired suppressive function. Mechanistically, CXXC1 interacts with Foxp3 and regulates the expression of key Treg signature genes by modulating H3K4me3 deposition. Their findings are interesting and significant. However, there are several concerns regarding their analysis and conclusions.

Major concerns:

(1) Despite cKO mice showing an increase in Treg cells in the lymph nodes and Cxxc1-deficient Treg cells having normal suppressive function, the majority of cKO mice died within a month. What causes cKO mice to die from severe inflammation?

Considering the results of Figures 4 and 5, a decrease in Treg cell population due to their reduced proliferative capacity may be one of the causes. It would be informative to analyze the population of tissue Treg cells.

(2) In Figure 5B, scRNA-seq analysis indicated that Mki67 $^{+}$ Treg subset are comparable between WT and Cxxc1-deficient Treg cells. On the other hand, FACS analysis demonstrated that Cxxc1-deficient Treg shows less Ki-67 expression compared to WT in Figure 5I. The authors should explain this discrepancy.

In addition, the authors concluded on line 441 that CXXC1 plays a crucial role in maintaining Treg cell stability. However, there appears to be no data on Treg stability. Which data represent the Treg stability?

(3) The authors found that *Cxxc1*-deficient Treg cells exhibit weaker H3K4me3 signals compared to WT in Figure 7. This result suggests that *Cxxc1* regulates H3K4me3 modification via H3K4 methyltransferases in Treg cells. The authors should clarify which H3K4 methyltransferases contribute to the modulation of H3K4me3 deposition by *Cxxc1* in Treg cells.

Furthermore, it would be important to investigate whether *Cxxc1*-deletion alters Foxp3 binding to target genes.

(4) In Figure 7, the authors concluded that CXXC1 promotes Treg cell homeostasis and function by preserving the H3K4me3 modification since *Cxxc1*-deficient Treg cells show lower H3K4me3 densities at the key Treg signature genes. Are these *Cxxc1*-deficient Treg cells derived from mosaic mice? If *Cxxc1*-deficient Treg cells are derived from cKO mice, the gene expression and H3K4me3 modification status are inconsistent because scRNA-seq analysis indicated that expression of these Treg signature genes was increased in *Cxxc1*-deficient Treg cells compared to WT (Figure 5F and G).

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Author response:

Reviewer #1 (Public review):

Summary:

This work investigated the role of CXXC-finger protein 1 (CXXC1) in regulatory T cells. CXXC1-bound genomic regions largely overlap with Foxp3-bound regions and regions with H3K4me3 histone modifications in Treg cells. CXXC1 and Foxp3 interact with each other, as shown by co-immunoprecipitation. Mice with Treg-specific CXXC1 knockout (KO) succumb to lymphoproliferative diseases between 3 to 4 weeks of age, similar to Foxp3 KO mice. Although the immune suppression function of CXXC1 KO Treg is comparable to WT Treg in an in vitro assay, these KO Tregs failed to suppress autoimmune diseases such as EAE and colitis in Treg transfer models in vivo. This is partly due to the diminished survival of the KO Tregs after transfer. CXXC1 KO Tregs do not have an altered DNA methylation pattern; instead, they display weakened H3K4me3 modifications within the broad H3K4me3 domains, which contain a set of Treg signature genes. These results suggest that CXXC1 and Foxp3 collaborate to regulate Treg homeostasis and function by promoting Treg signature gene expression through maintaining H3K4me3 modification.

Strengths:

Epigenetic regulation of Treg cells has been a constantly evolving area of research. The current study revealed CXXC1 as a previously unidentified epigenetic regulator of Tregs. The strong phenotype of the knockout mouse supports the critical role CXXC1 plays in Treg cells. Mechanistically, the link between CXXC1 and the maintenance of broad H3K4me3 domains is also a novel finding.

Weaknesses:

(1) It is not clear why the authors chose to compare H3K4me3 and H3K27me3 enriched genomic regions. There are other histone modifications associated with transcription activation or repression. Please provide justification.

Thank you for highlighting this important point. We prioritized H3K4me3 and H3K27me3 because they are well-established markers of transcriptional activation and repression, respectively. These modifications provide a robust framework for investigating the dynamic interplay of chromatin states in Treg cells, particularly in regulating the balance between activation and suppression of key genes. While histone acetylation, such as H3K27ac, is linked to enhancer activity and transcriptional elongation, our focus was on promoter-level regulation, where H3K4me3 and H3K27me3 are most relevant. Although other histone modifications could provide additional insights, we chose to focus on these two to maintain clarity and feasibility in our analysis. We are happy to further elaborate on this rationale in the manuscript if necessary.

(2) It is not clear what separates Clusters 1 and 3 in Figure 1C. It seems they share the same features.

We apologize for not clarifying these clusters clearly. Cluster 1 and 3 are both H3K4me3 only group, with H3K4me3 enrichment and gene expression levels being higher in Cluster 1. At first, we divided the promoters into four categories because we wanted to try to classify them into four categories: H3K4me3 only, H3K27me3 only, H3K4me3-H3K27me3 co-occupied, and None. However, in actual classification, we could not distinguish H3K4me3-H3K27me3 co-occupied group. Instead, we had two categories of H3K4me3 only, with cluster 1 having a higher enrichment level for H3K4me3 and gene expression levels.

(3) The claim, "These observations support the hypothesis that FOXP3 primarily functions as an activator by promoting H3K4me3 deposition in Treg cells." (line 344), seems to be a bit of an overstatement. Foxp3 certainly can promote transcription in ways other than promoting H3K3me3 deposition, and it also can repress gene transcription without affecting H3K27me3 deposition. Therefore, it is not justified to claim that promoting H3K4me3 deposition is Foxp3's primary function.

We appreciate the reviewer's thoughtful observation regarding our claim about FOXP3's role in promoting H3K4me3 deposition. We acknowledge that FOXP3 is a multifunctional transcription factor with diverse mechanisms of action, including transcriptional activation independent of H3K4me3 deposition and transcriptional repression that does not necessarily involve H3K27me3 deposition.

Our intention was not to imply that promoting H3K4me3 deposition is the exclusive or predominant function of FOXP3 but rather to highlight that this mechanism contributes significantly to its role in regulating Treg cell function. We agree that our wording may have overstated this point, and we will revise the text to provide a more nuanced interpretation. Specifically, we will clarify that our observations suggest FOXP3 can facilitate transcriptional activation, in part, by promoting H3K4me3 deposition, but this does not preclude its other regulatory mechanisms.

(4) For the in vitro suppression assay in Figure S4C, and the Treg transfer EAE and colitis experiments in Figure 4, the Tregs should be isolated from Cxhc1 fl/fl x Foxp3 cre/wt female heterozygous mice instead of Cxhc1 fl/fl x Foxp3 cre/cre (or cre/Y) mice. Tregs from the homozygous KO mice are already activated by the lymphoproliferative environment and could have vastly different gene expression patterns and homeostatic features compared to resting Tregs. Therefore, it's not a fair comparison between these activated KO Tregs and resting WT Tregs.

Thank you for this insightful comment and for pointing out the potential confounding effects associated with using Treg cells from homozygous Foxp3Cre/Cre (or Cre/Y) Cxhc1fl/fl mice. We agree that using Treg cells from *_Foxp3_Cre/+_Cxhc1_fl/fl* (referred to as "het-KO") and their

littermate *Foxp3*^{Cre/+} *Cxhc1*^{fl/+} (referred to as “het-WT”) female mice would provide a more balanced comparison, as these Treg cells are less likely to be influenced by the activated lymphoproliferative environment present in homozygous KO mice.

To address this concern, we will perform additional experiments using Treg cells isolated from *Foxp3*^{Cre/+} *Cxhc1*^{fl/fl} (“het-KO”) and their littermate *Foxp3*^{Cre/+} *Cxhc1*^{fl/+} (“het-WT”) female mice. We will update the manuscript with these new data to provide a more accurate assessment of the impact of CXXC1 deficiency on Treg cell function.

(5) The manuscript didn't provide a potential mechanism for how CXXC1 strengthens broad H3K4me3-modified genomic regions. The authors should perform Foxp3 ChIP-seq or Cut-n-Tag with WT and Cxhc1 cKO Tregs to determine whether CXXC1 deletion changes Foxp3's binding pattern in Treg cells.

Thank you for your insightful comments and valuable suggestions. We greatly appreciate your recommendation to explore the potential mechanism by which CXXC1 enhances broad H3K4me3-modified genomic regions.

In response, we plan to conduct CUT&Tag experiments for Foxp3 in both WT and *Cxhc1* cKO Treg cells.

Reviewer #2 (Public review):

FOXP3 has been known to form diverse complexes with different transcription factors and enzymes responsible for epigenetic modifications, but how extracellular signals timely regulate FOXP3 complex dynamics remains to be fully understood. Histone H3K4 tri-methylation (H3K4me3) and CXXC finger protein 1 (CXXC1), which is required to regulate H3K4me3, also remain to be fully investigated in Treg cells. Here, Meng et al. performed a comprehensive analysis of H3K4me3 CUT&Tag assay on Treg cells and a comparison of the dataset with the FOXP3 ChIP-seq dataset revealed that FOXP3 could facilitate the regulation of target genes by promoting H3K4me3 deposition.

Moreover, CXXC1-FOXP3 interaction is required for this regulation. They found that specific knockdown of Cxhc1 in Treg leads to spontaneous severe multi-organ inflammation in mice and that Cxhc1-deficient Treg exhibits enhanced activation and impaired suppression activity. In addition, they have also found that CXXC1 shares several binding sites with FOXP3 especially on Treg signature gene loci, which are necessary for maintaining homeostasis and identity of Treg cells.

The findings of the current study are pretty intriguing, and it would be great if the authors could fully address the following comments to support these interesting findings.

Major points:

(1) There is insufficient evidence in the first part of the Results to support the conclusion that "FOXP3 functions as an activator by promoting H3K4Me3 deposition in Treg cells". The authors should compare the results for H3K4Me3 in FOXP3-negative conventional T cells to demonstrate that at these promoter loci, FOXP3 promotes H3K4Me3 deposition.

We appreciate the reviewer’s critical observation regarding our claim about FOXP3’s role in promoting H3K4me3 deposition. We acknowledge that FOXP3 is a multifunctional transcription factor with diverse mechanisms of action, including transcriptional activation independent of H3K4me3 deposition and transcriptional repression that does not necessarily involve H3K27me3 deposition.

Our intention was not to imply that promoting H3K4me3 deposition is the exclusive or predominant function of FOXP3 but rather to highlight that this mechanism contributes

significantly to its role in regulating Treg cell function. We agree that our wording may have overstated this point, and we will revise the text to provide a more nuanced interpretation. Specifically, we will clarify that our observations suggest FOXP3 can facilitate transcriptional activation, in part, by promoting H3K4me3 deposition, but this does not preclude its other regulatory mechanisms.

We will compare H3K4me3 levels at the promoter loci of interest between FOXP3-negative conventional T cells and FOXP3-positive regulatory T cells. This comparison will help elucidate whether FOXP3 directly promotes H3K4me3 deposition at these loci.

(2) In Figure 3 F&G, the activation status and IFN γ production should be analyzed in Treg cells and Tconv cells separately rather than in total CD4 $^{+}$ T cells. Moreover, are there changes in autoantibodies and IgG and IgE levels in the serum of cKO mice?

We appreciate the reviewer's constructive feedback on the analyses presented in Figures 3F and 3G and the additional suggestion to investigate autoantibodies and serum immunoglobulin levels.

Regarding Figures 3F and 3G, we agree that separating Treg cells and Tconv cells for analysis of activation status and IFN- γ production would provide a more precise understanding of the cellular dynamics in *Cxxc1* cKO mice.

To address this, we will reanalyze the data to examine Treg and Tconv cells independently and include these results in the revised manuscript.

As for the changes in autoantibodies and serum IgG and IgE levels, we acknowledge that these parameters are important indicators of systemic immune dysregulation.

We will now measure serum autoantibodies and immunoglobulin levels in *Cxxc1* cKO mice and WT controls.

(3) Why did Cxxc1-deficient Treg cells not show impaired suppression than WT Treg during in vitro suppression assay, despite the reduced expression of Treg cell suppression assay-associated markers at the transcriptional level demonstrated in both scRNA-seq and bulk RNA-seq?

Thank you for your thoughtful question. We appreciate your interest in understanding the apparent discrepancy between the reduced expression of Treg-associated suppression markers at the transcriptional level and the lack of impaired suppression observed in the in vitro suppression assay.

There are several potential explanations for this observation:

(1) Functional Redundancy: Treg cell suppression is a complex, multi-faceted process involving various effector mechanisms such as cytokine production (e.g., IL-10, TGF- β), cell-cell contact, and metabolic regulation. Thus, even though the transcriptional signature of suppression-associated genes is altered, compensatory mechanisms may still allow *Cxxc1*-deficient Treg cells to retain functional suppression capacity under these specific in vitro conditions.

(2) In Vitro Assay Limitations: The in vitro suppression assay is a simplified model of Treg function that may not capture all the complexities of Treg-mediated suppression in vivo. While we observed altered gene expression in *Cxxc1*-deficient Treg cells, this might not directly translate to a functional defect under the specific conditions of the assay. In vivo, additional factors such as cytokine milieu, cell-cell interactions, and tissue-specific environments may be required for full suppression, which could be missing in the in vitro assay.

(4) Is there a disease in which *Cxhc1* is expressed at low levels or absent in Treg cells? Is the same immunodeficiency phenotype present in patients as in mice?

Thank you for your insightful question regarding the role of CXXC1 in Treg cells and its potential link to human disease. To our knowledge, no specific human disease has been identified where CXXC1 is expressed at low levels or absent specifically in Treg cells. There is currently no direct evidence of an immunodeficiency phenotype in human patients that parallels the one observed in *Cxhc1*-deficient mice.

Reviewer #3 (Public review):

In the report entitled "CXXC-finger protein 1 associates with FOXP3 to stabilize homeostasis and suppressive functions of regulatory T cells", the authors demonstrated that Cxhc1-deletion in Treg cells leads to the development of severe inflammatory disease with impaired suppressive function. Mechanistically, CXXC1 interacts with Foxp3 and regulates the expression of key Treg signature genes by modulating H3K4me3 deposition. Their findings are interesting and significant. However, there are several concerns regarding their analysis and conclusions.

Major concerns:

(1) Despite cKO mice showing an increase in Treg cells in the lymph nodes and Cxhc1-deficient Treg cells having normal suppressive function, the majority of cKO mice died within a month. What causes cKO mice to die from severe inflammation?

Considering the results of Figures 4 and 5, a decrease in Treg cell population due to their reduced proliferative capacity may be one of the causes. It would be informative to analyze the population of tissue Treg cells.

We thank the reviewer for this insightful comment and acknowledge the importance of understanding the causes of severe inflammation and early mortality in cKO mice. Based on our data and previous studies, we propose the following explanations:

(1) Reduced Treg Proliferative Capacity: As shown in Figure 5I, the decreased proportion of FOXP3+Ki67+ Treg cells in cKO mice likely reflects impaired proliferative capacity, which may limit the expansion of functional Treg cells in response to inflammatory cues, particularly in peripheral tissues where active suppression is required.

(2) Altered Treg Function and Activation: *Cxhc1*-deficient Treg cells exhibit increased expression of activation markers (*Il2ra*, *Cd69*) and pro-inflammatory genes (*Ifng*, *Tbx21*). This suggests a functional dysregulation that may impair their ability to suppress inflammation effectively, despite their presence in lymphoid organs.

(3) Tissue Treg Populations: Although our study focuses on lymph node-resident Treg cells, tissue-resident Treg cells play a crucial role in maintaining local immune homeostasis. It is plausible that *Cxhc1* deficiency compromises the accumulation or functionality of tissue Treg cells, contributing to uncontrolled inflammation in non-lymphoid organs. Unfortunately, we currently lack data on tissue Treg populations, which limits our ability to directly address this hypothesis.

Regarding the suggestion to analyze tissue Treg populations, we agree that this would be an important next step in understanding the cause of the severe inflammation and early mortality in *Cxhc1*-deficient mice.

We plan to perform detailed analyses of Treg cell populations in various tissues, including the gut, lung, and liver, to determine if there are specific defects in tissue-resident Treg cells that

could contribute to the observed phenotype.

(2) In Figure 5B, scRNA-seq analysis indicated that Mki67+ Treg subset are comparable between WT and Cxxc1-deficient Treg cells. On the other hand, FACS analysis demonstrated that Cxxc1-deficient Treg shows less Ki-67 expression compared to WT in Figure 5I. The authors should explain this discrepancy.

Thank you for pointing out the apparent discrepancy between the scRNA-seq and FACS analyses regarding Ki-67 expression in *Cxxc1*-deficient Treg cells.

In Figure 5B, the scRNA-seq analysis identified the Mki67+ Treg subset as comparable between WT and *Cxxc1*-deficient Treg cells. This finding reflects the overall proportion of cells expressing Mki67 transcripts within the Treg population. In contrast, the FACS analysis in Figure 5I specifically measures Ki-67 protein levels, revealing reduced expression in *Cxxc1*-deficient Treg cells compared to WT.

To address this discrepancy more comprehensively, we will further analyze the scRNA-seq data to directly compare Mki67 mRNA expression levels between WT and *Cxxc1*-deficient Treg cells.

In addition, the authors concluded on line 441 that CXXC1 plays a crucial role in maintaining Treg cell stability. However, there appears to be no data on Treg stability. Which data represent the Treg stability?

We appreciate the reviewer's observation and recognize that our wording may have been overly conclusive. Our data primarily highlight the impact of *Cxxc1* deficiency on Treg cell homeostasis and transcriptional regulation, rather than providing direct evidence for Treg cell stability. Specifically, the downregulation of Treg-specific suppressive genes (*Nt5e*, *Il10*, *Pdcd1*) and the upregulation of pro-inflammatory markers (*Gzmb*, *Ifng*, *Tbx21*) indicate a shift in functional states. While these findings may suggest an indirect disruption in the maintenance of suppressive phenotypes, they do not constitute a direct measure of Treg cell stability.

To address the reviewer's concern, we will revise our conclusion to more accurately state that our data support a role for CXXC1 in maintaining Treg cell homeostasis and functional balance, without overextending claims about Treg cell stability. Thank you for bringing this to our attention, as it will help us improve the clarity and precision of our manuscript.

(3) The authors found that Cxxc1-deficient Treg cells exhibit weaker H3K4me3 signals compared to WT in Figure 7. This result suggests that Cxxc1 regulates H3K4me3 modification via H3K4 methyltransferases in Treg cells. The authors should clarify which H3K4 methyltransferases contribute to the modulation of H3K4me3 deposition by Cxxc1 in Treg cells.

Thank you for pointing out the need to clarify the role of H3K4 methyltransferases in the modulation of H3K4me3 deposition by CXXC1 in Treg cells.

In our study, we found that *Cxxc1*-deficient Treg cells exhibit reduced H3K4me3 levels, as shown in Figure 7. CXXC1 has been previously reported to function as a non-catalytic component of the Set1/COMPASS complex, which contains H3K4 methyltransferases such as SETD1A and SETD1B. These methyltransferases are the primary enzymes responsible for H3K4 trimethylation.

References:

(1) Lee J.H., Skalnik D.G. CpG-binding protein (CXXC finger protein 1) is a component of the mammalian Set1 histone H3-Lys4 methyltransferase complex, the analogue of the yeast

Set1/COMPASS complex. J. Biol. Chem. 2005; 280:41725–41731.

(2). J. P. Thomson, P. J. Skene, J. Selfridge, T. Clouaire, J. Guy, S. Webb, A. R. W. Kerr, A. Deaton, R. Andrews, K. D. James, D. J. Turner, R. Illingworth, A. Bird, CpG islands influence chromatin structure via the CpG-binding protein Cfp1. Nature 464, 1082–1086 (2010).

(3) Shilatifard, A. 2012. The COMPASS family of histone H3K4 methylases: mechanisms of regulation in development and disease pathogenesis. Annu. Rev. Biochem. 81:65–95.

(4) Brown D.A., Di Cerbo V., Feldmann A., Ahn J., Ito S., Blackledge N.P., Nakayama M., McClellan M., Dimitrova E., Turberfield A.H. et al. The SET1 complex selects actively transcribed target genes via multivalent interaction with CpG Island chromatin. Cell Rep. 2017; 20:2313–2327.

Furthermore, it would be important to investigate whether Cxxc1-deletion alters Foxp3 binding to target genes.

Thank you for this important suggestion regarding the impact of Cxxc1 deletion on FOXP3 binding to target genes. We agree that understanding whether Cxxc1 deficiency affects FOXP3's ability to bind to its target genes would provide valuable insight into the regulatory role of CXXC1 in Treg cell function.

To address this, we plan to perform CUT&Tag experiments to assess FOXP3 binding profiles in Cxxc1-deficient versus wild-type Treg cells. These experiments will allow us to determine if Cxxc1 loss disrupts FOXP3's occupancy at key regulatory sites, which may contribute to the observed functional impairments in Treg cells.

(4) In Figure 7, the authors concluded that CXXC1 promotes Treg cell homeostasis and function by preserving the H3K4me3 modification since Cxxc1-deficient Treg cells show lower H3K4me3 densities at the key Treg signature genes. Are these Cxxc1-deficient Treg cells derived from mosaic mice? If Cxxc1-deficient Treg cells are derived from cKO mice, the gene expression and H3K4me3 modification status are inconsistent because scRNA-seq analysis indicated that expression of these Treg signature genes was increased in Cxxc1-deficient Treg cells compared to WT (Figure 5F and G).

Thank you for the insightful comment. To clarify, the Cxxc1-deficient Treg cells analyzed for H3K4me3 modification in Figure 7 were indeed derived from Cxxc1 conditional knockout (cKO) mice, not mosaic mice.

The scRNA-seq analysis presented in Figures 5F and G revealed an upregulation of Treg signature genes in Cxxc1-deficient Treg cells. This finding suggests that the loss of Cxxc1 drives these cells toward a pro-inflammatory, activated state, underscoring the pivotal role of CXXC1 in maintaining Treg cell homeostasis and suppressive function.

Regarding the apparent discrepancy between the reduced H3K4me3 levels and the increased expression of these genes, it is important to note that H3K4me3 primarily functions as an epigenetic mark that facilitates chromatin accessibility and transcriptional regulation, acting as an upstream modulator of gene expression. However, gene expression levels are also influenced by downstream compensatory mechanisms and complex inflammatory environments. In this context, the reduction in H3K4me3 likely reflects the direct role of CXXC1 in epigenetic regulation, whereas the upregulation of gene expression in Cxxc1-deficient Treg cells may result as a side effect of the inflammatory environment.

To further substantiate our findings, we performed RNA-seq analysis on Treg cells from *_Foxp3_Cre/+ _Cxxc1^{fl/fl}* (“het-KO”) and their littermate *_Foxp3_Cre/+ _Cxxc1^{fl/+}* (“het-WT”) female mice, as presented in Figure S6C. This analysis revealed a notable reduction in the

expression of key Treg signature genes, including *Icos*, *Ctla4*, *Tnfrsf18*, and *Nt5e*, in het-KO Treg cells. Importantly, the observed changes in gene expression were consistent with the altered H3K4me3 modification status, further supporting the epigenetic regulatory role of CXXC1. These results further emphasize the critical role of CXXC1 promotes Treg cell homeostasis and function by preserving the H3K4me3 modification.

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